

Mixed models

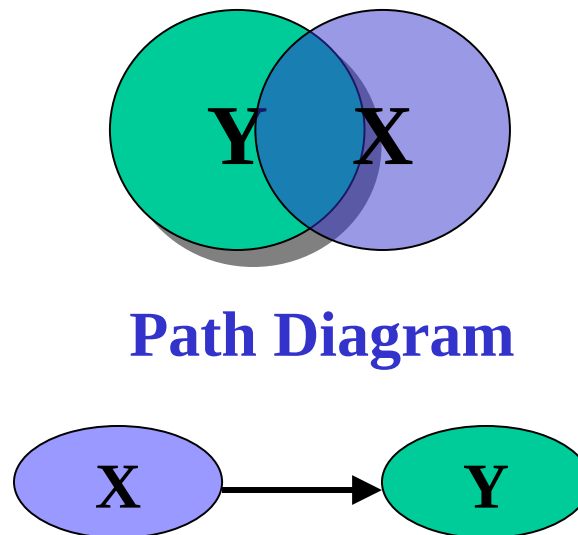
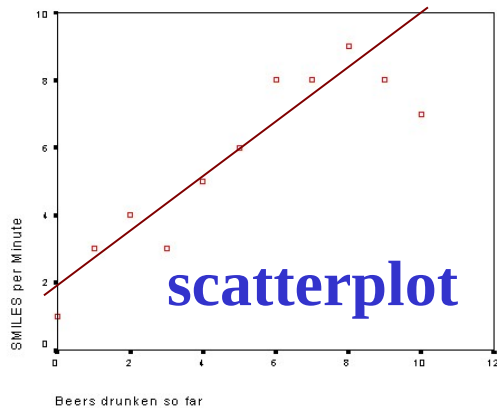
Part I



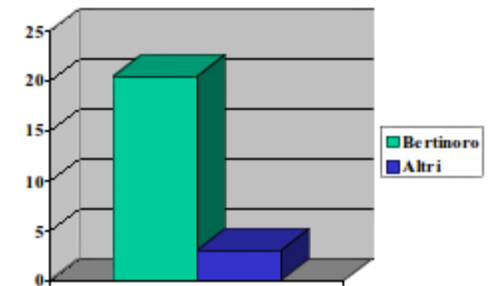
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Introduction

A simple **statistical model** is an efficient and concise representation of the data describing an empirical phenomenon



Difference in mean

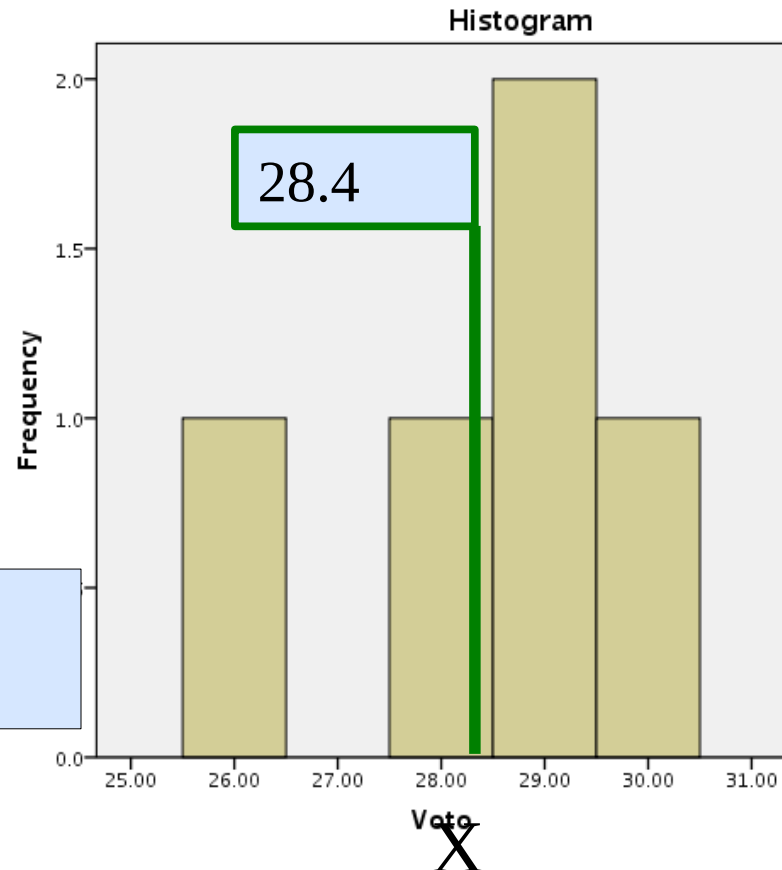


Introduction

The statistical model and its representation is aimed at:

- Efficient and Concise Description
- Prediction of future
- Generalization to a population

That is: understanding of main trends in data



GLM

- The majority of the statistical technique that we use belong to one single **general model**

General Linear Model

$$y_i = a + b_1 \cdot x_{1i} + b_2 \cdot x_{2i} + \dots + b_k \cdot x_{ki} + e_i$$

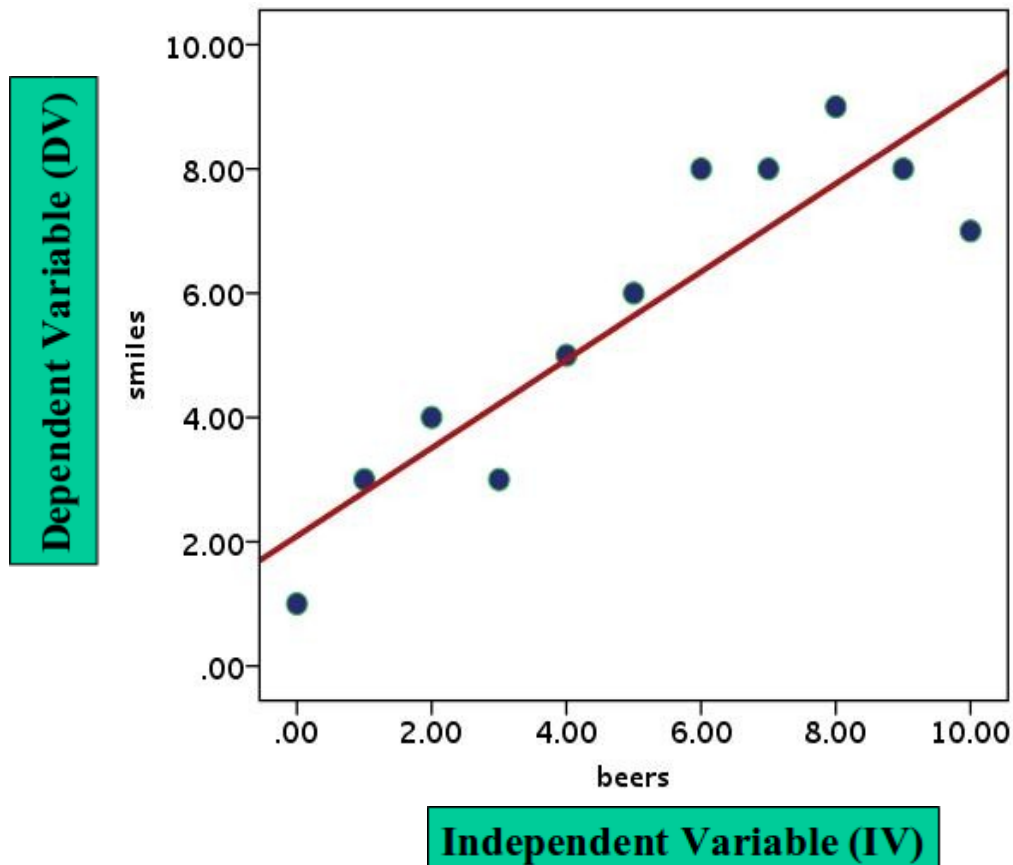
**Dependent
variable**

Independent variables

Errors

GLM Example

The aim of regression analysis is to fit the data using a function



For most applications, we just need a linear function: straight line

$$y_i = a + b \cdot x_i + e_i$$

$$\hat{y}_i = a + b \cdot x_i$$

GLM

pros

- It allows to estimate the effects of one or several IVs on a DV
- It can be applied in many different research problems
- It allows to estimate many different types of effects

cons

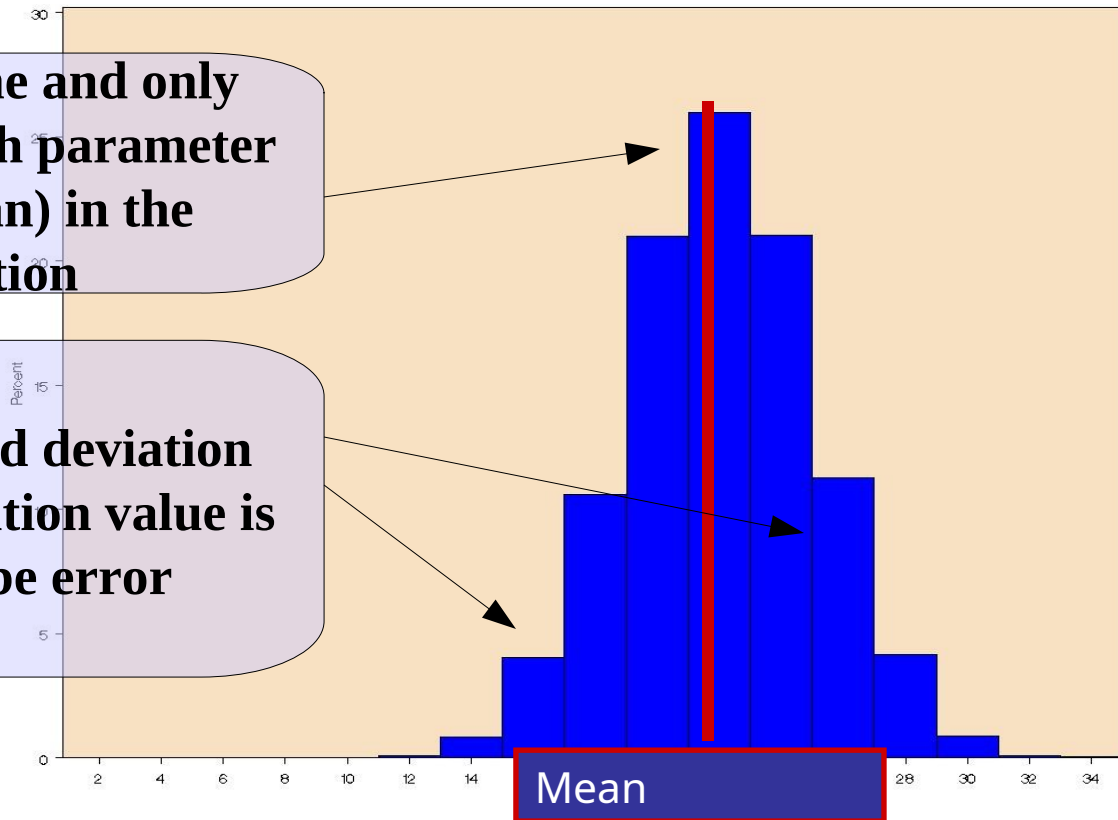
- It assumes data have a very simple structure
- It applies on a limited amount of dependent variables (continuous or semi-continuous DV)

GLM Assumptions

$$y_i = a + e_i$$

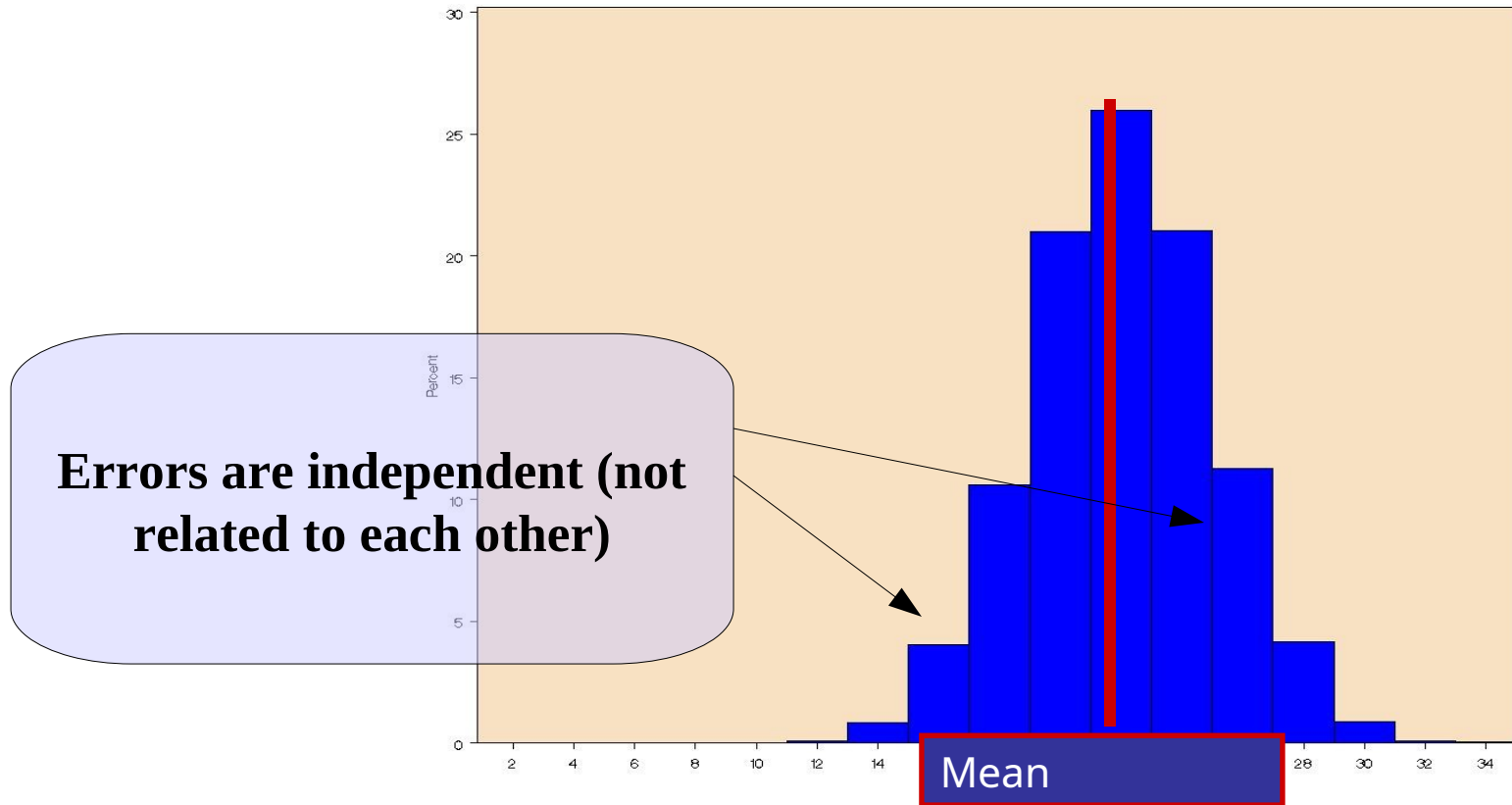
1) There exists one and only one value of each parameter (e.g. the mean) in the population

2) Any observed deviation from the population value is deemed to be error



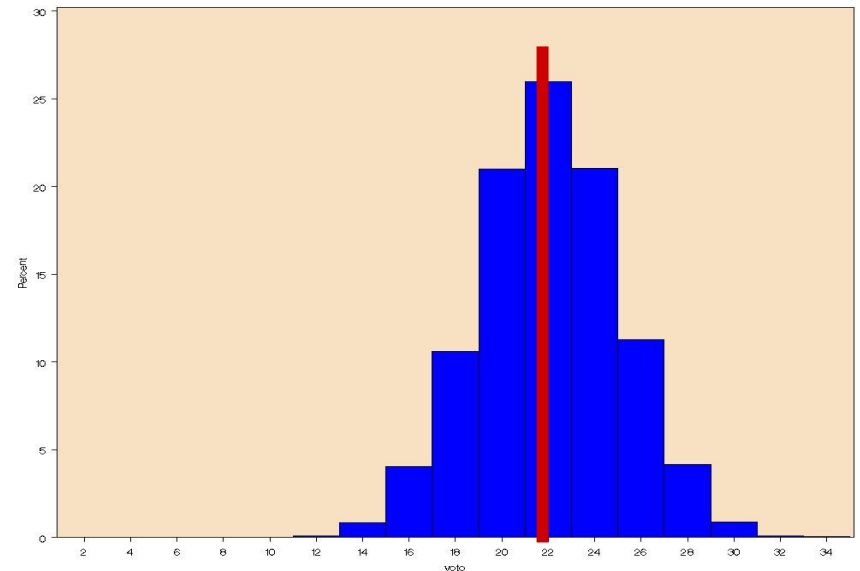
GLM Assumptions

$$y_i = a + e_i$$



GLM Assumptions

The estimated value is a
fixed parameter

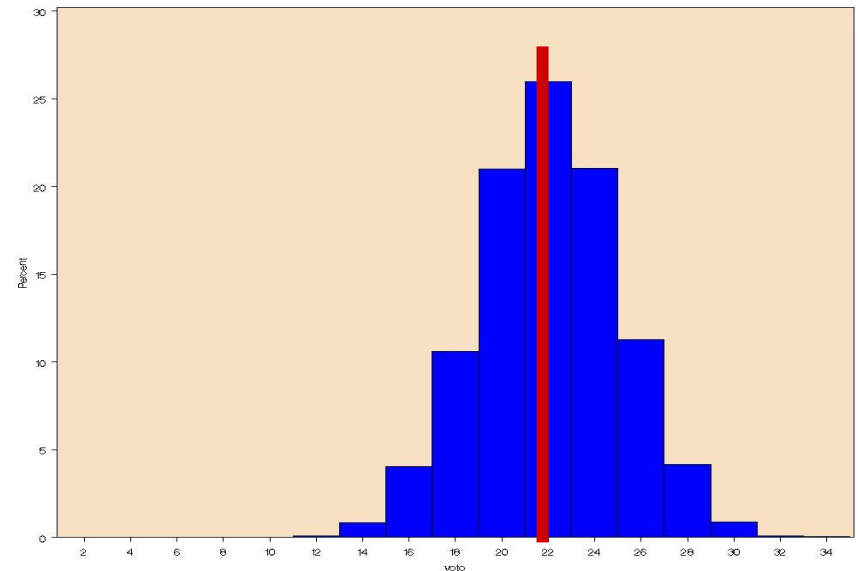


$$y_i = a + e_i$$
$$\text{corr}(e_i, e_j) = 0$$

Random variations are
uncorrelated

GLM Assumptions

The estimated value is a
fixed parameter



$$y_i = a + e_i$$
$$\text{corr}(e_i, e_j) = 0$$

Random variations are
uncorrelated

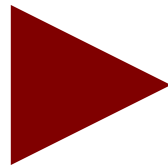
GLM

When the assumptions are NOT met because the data, and thus the errors have more complex structures, we generalize the GLM to the Linear Mixed Model

Linear Mixed Model

GLM

Regression
T-test
ANOVA
ANCOVA
Moderation
Mediation
Path Analysis



LMM

Random coefficients models
Random intercept regression models
One-way ANOVA with random effects
One-way ANCOVA with random effects
Intercepts-and-slopes-as-outcomes models
Multi-level models

Mixed Linear Models

- With the mixed model one can take into the account dependency among measures (within clusters) almost in any situation
- It allows applying the GLM logic to a broader range of designs
- Any kind of independent variables
- Efficient handling of missing values
- Multi-level research designs
- Repeated measures designs
- Generalizes to the generalized linear model (logistic etc)

Example “beers”

Let's consider the case where the beer-smile research was conducted by gathering data in several different bars

For each participant
we measured # of
beers and # of smiles

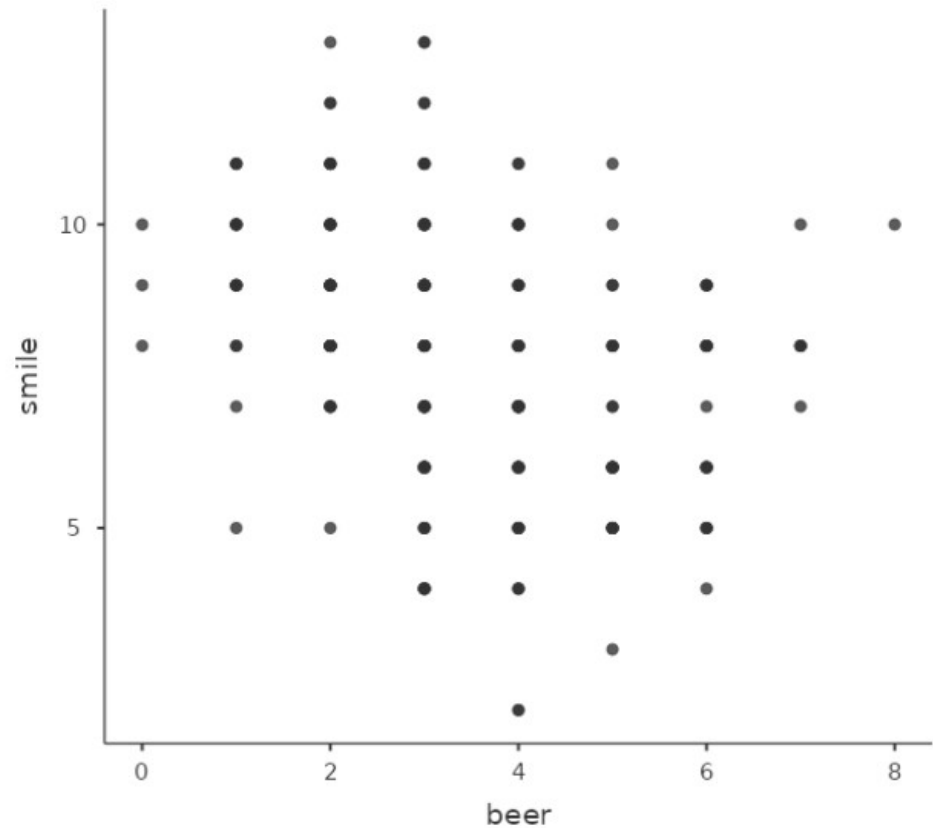
		bar			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	a	3	1.3	1.3	1.3
	b	14	6.0	6.0	7.3
	c	22	9.4	9.4	16.7
	d	21	9.0	9.0	25.6
	e	14	6.0	6.0	31.6
	f	20	8.5	8.5	40.2
	g	24	10.3	10.3	50.4
	h	12	5.1	5.1	55.6
	i	16	6.8	6.8	62.4
	l	22	9.4	9.4	71.8
	m	21	9.0	9.0	80.8
	n	15	6.4	6.4	87.2
	o	16	6.8	6.8	94.0
	p	11	4.7	4.7	98.7
	q	3	1.3	1.3	100.0
	Total	234	100.0	100.0	

For a total of 234 participants

Example “beers” 2

As compared with the example with a few participants, now we have a very different scatterplot

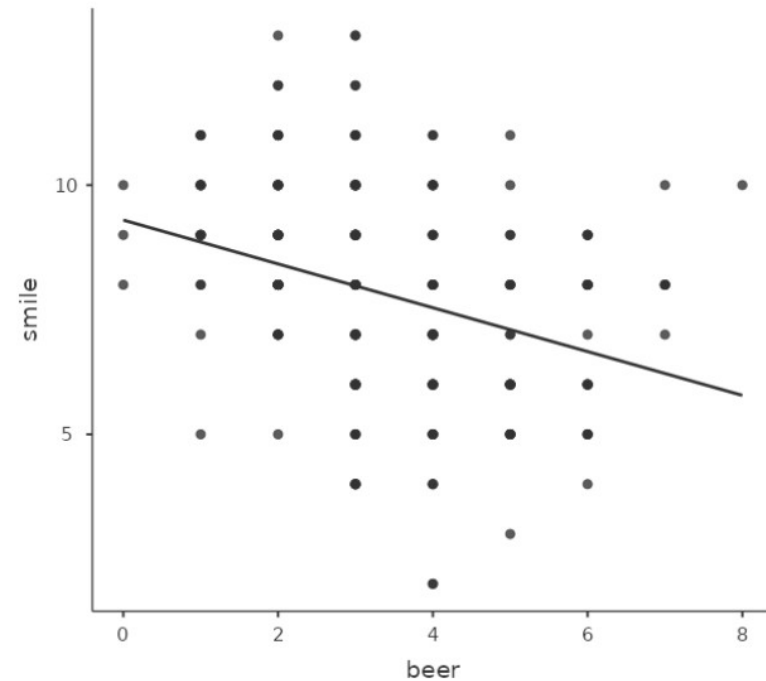
Scatterplot



Example “beers” 2

A simple regression confirms that results are indeed different

Scatterplot



Negative effect

Fixed Effects Parameter Estimates

Names	Estimate	SE	95% Confidence Interval		β	df	t	p
			Lower	Upper				
(Intercept)	7.765	0.130	7.508	8.022	0.000	232	59.503	< .001
beer	-0.440	0.085	-0.608	-0.271	-0.320	232	-5.147	< .001

Why

Results may be biased by a mis-specification of the model, where the structure of the data is not taken into account

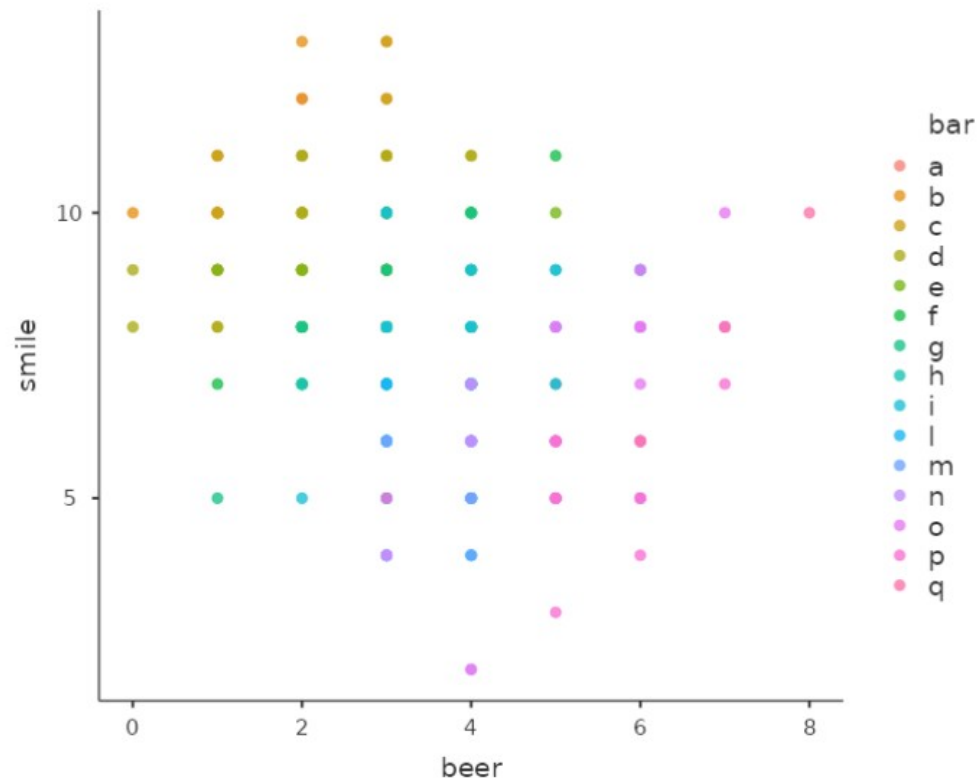
- In fact:
 - Subjects are sampled in clusters specified by **bars**
 - Each bar may have specific characteristics (quality, entertainment, etc) that may affect the measured variables
 - Subjects within the same bar may be more similar than across bars

Scatterplot by Bar

Let's see the data broken down by bar

Bar

Scatterplot

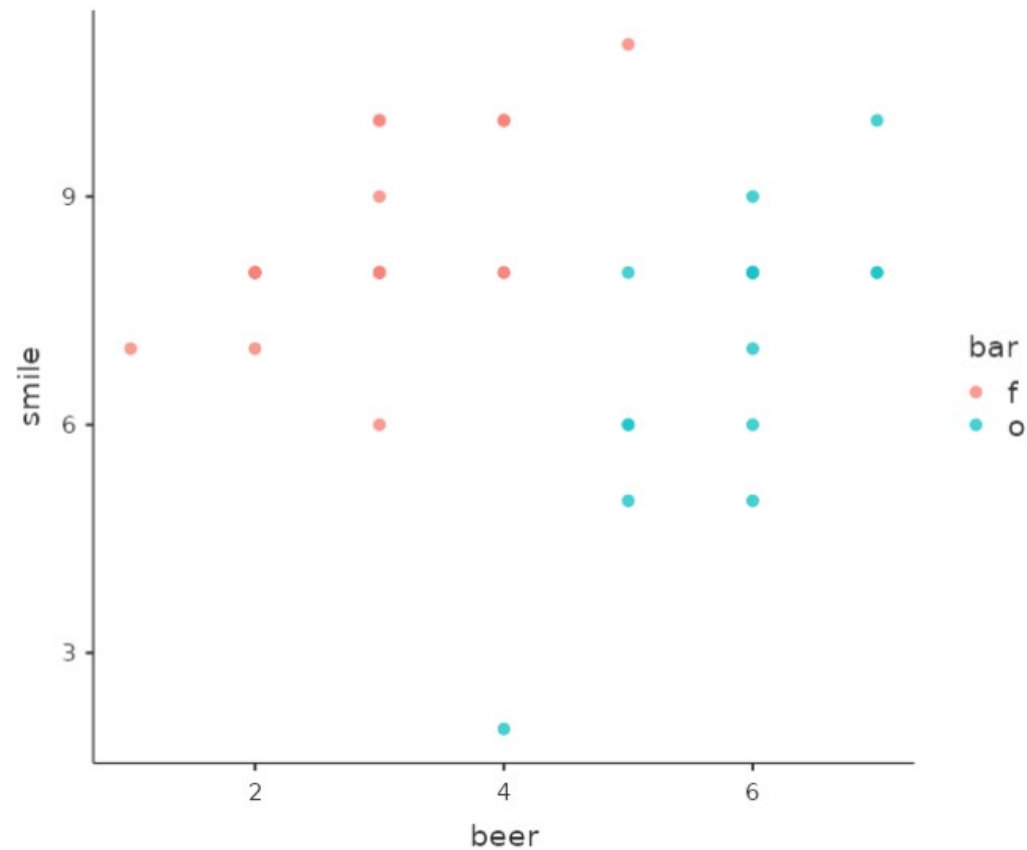


Scatterplot by Bar

Let's see the data only for bar “f” and “o”

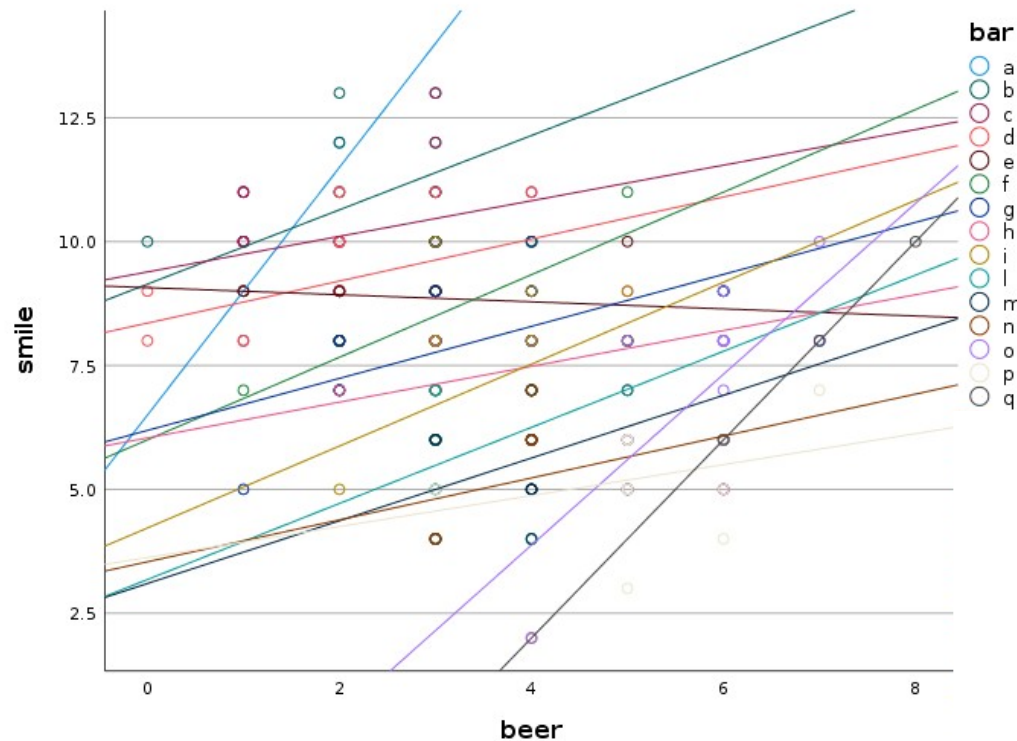
Bar

Scatterplot



Scatterplot by Bar

It seems that the relations between IV and DV is positive, but within each bar

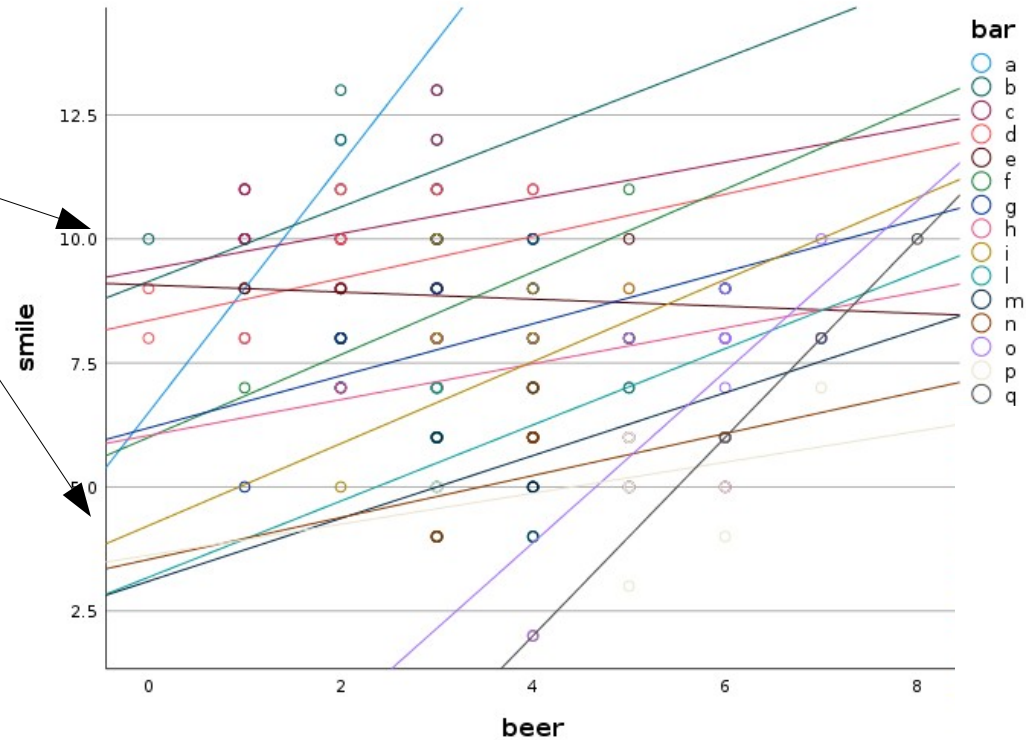


Bar

Scatterplot by Bar

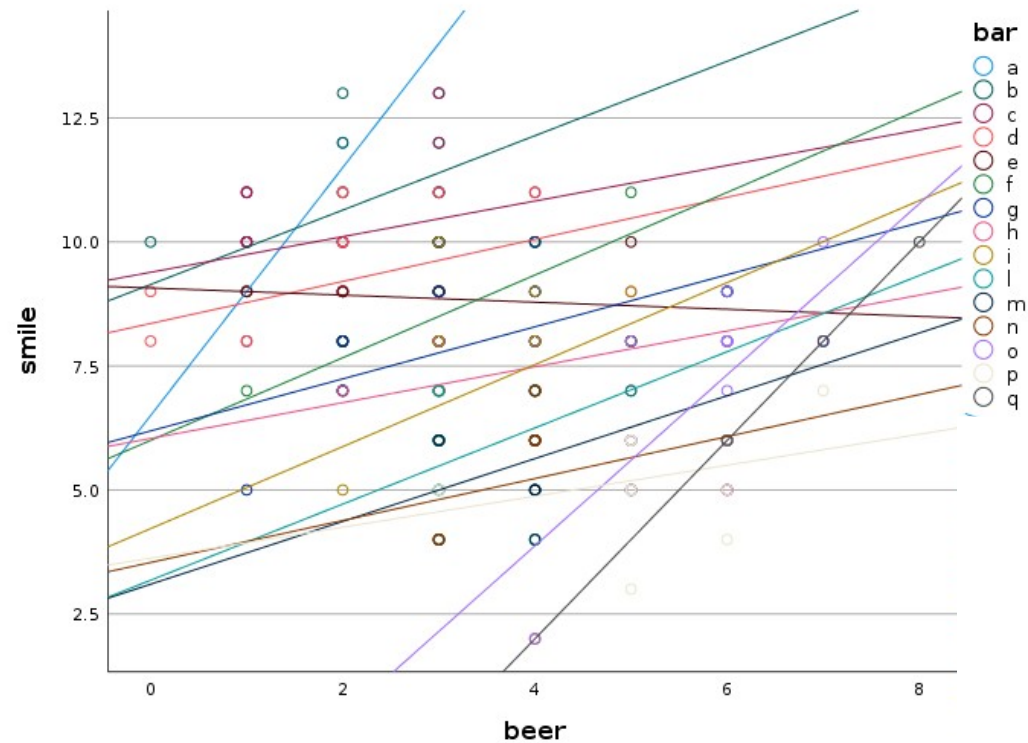
Bar

Intercepts seem to
be different from
bar to bar



Scatterplot by Bar

Bar



Slopes are all positive

Slopes seem to vary across bars

The Model

- It seems that considering the participants as all equivalent and independent one each other (GLM assumption) does not fit our data
- It seems that a better model should allow each bar (each cluster) to have a different regression line (a different intercept and **b** coefficient)

The Model

- Let's define a model with a regression line for each cluster

 y_{ij}

Smiles of subject i in the cluster j

$$\hat{y}_{ia} = a_a + b_a \cdot x_{ia}$$

$$\hat{y}_{ib} = a_b + b_b \cdot x_{ib}$$

$$\hat{y}_{ic} = a_c + b_c \cdot x_{ic}$$

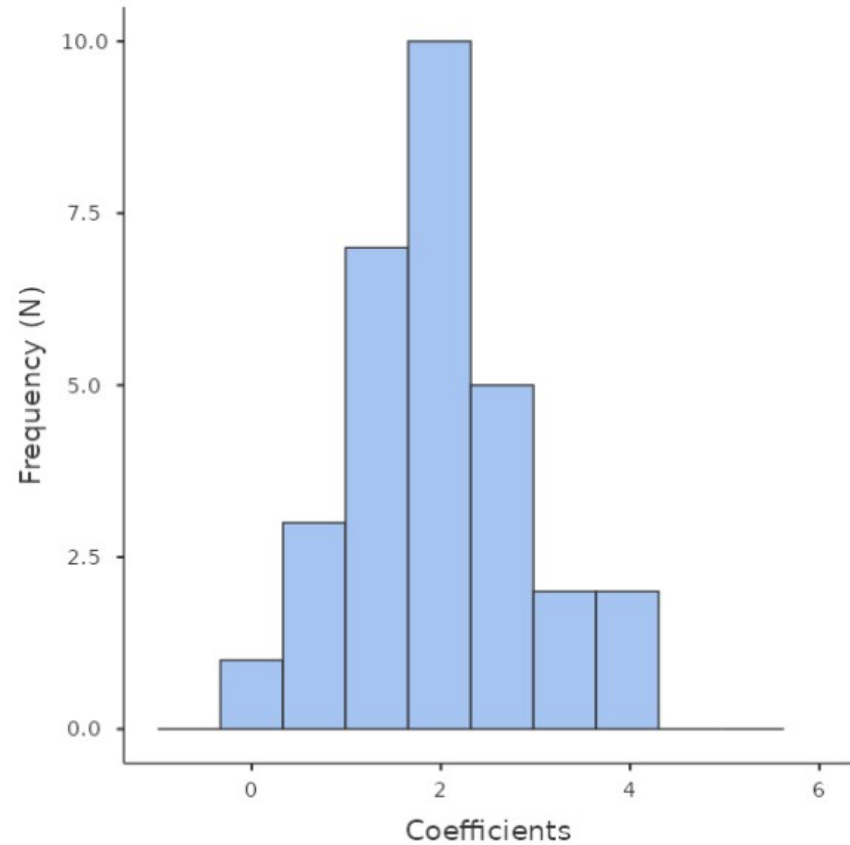
$$\hat{y}_{ij} = a_j + b_j \cdot x_{ij}$$

In these regressions the coefficients may vary from cluster to cluster: they are not **Fixed**

Varying coefficients

- If coefficients may vary, they will have a distribution

A possible distribution of coefficients b estimated for different clusters

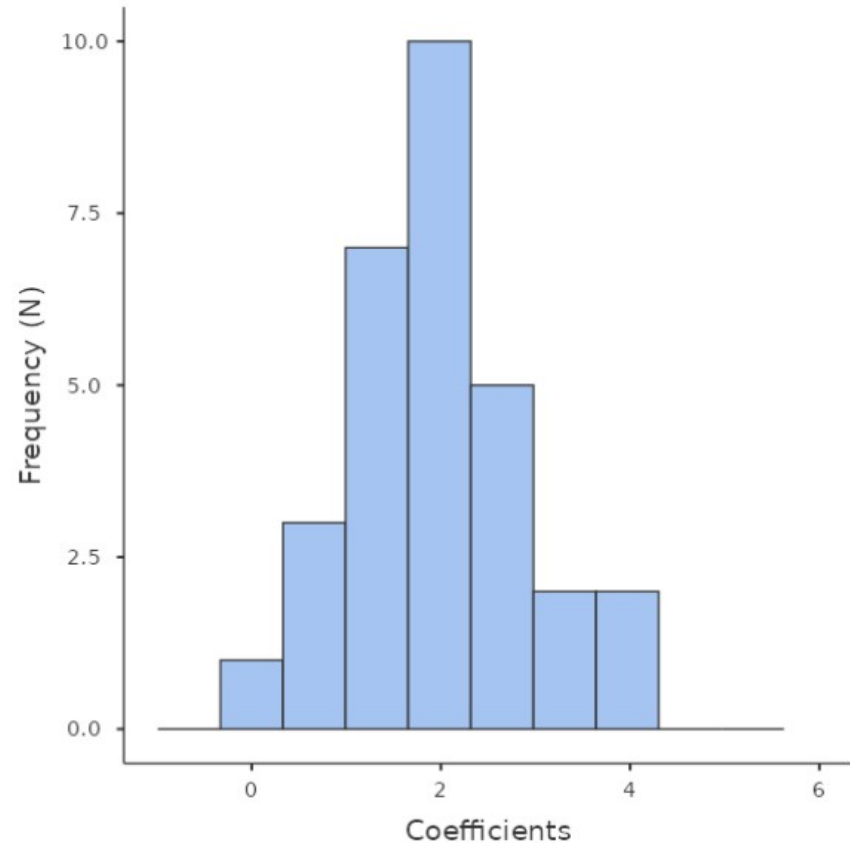


Random coefficients

- Varying coefficients are called **random coefficients**

Coefficients will exhibit variability: They are **random coefficients**

That is: in the **population** there exist different coefficients, a sample of which we estimated using the clustered data

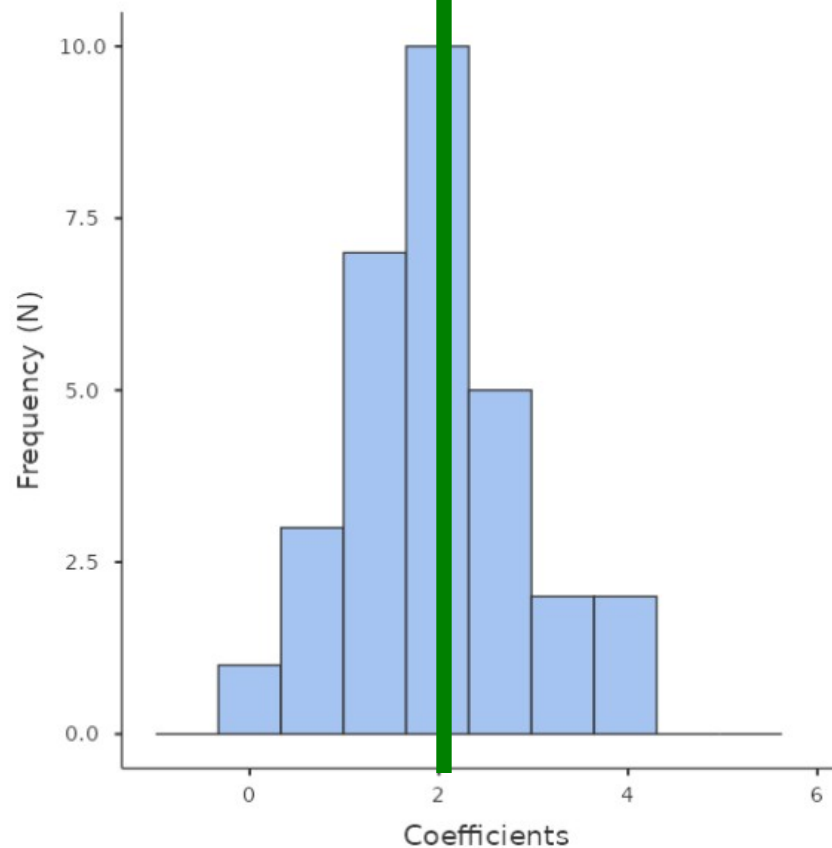


Average of the coefficients

- If coefficients vary as a variable in the population, they will have a mean and a variance, that we can estimate in our data

Mean

$$\bar{b} = \frac{\sum_j b_j}{k}$$



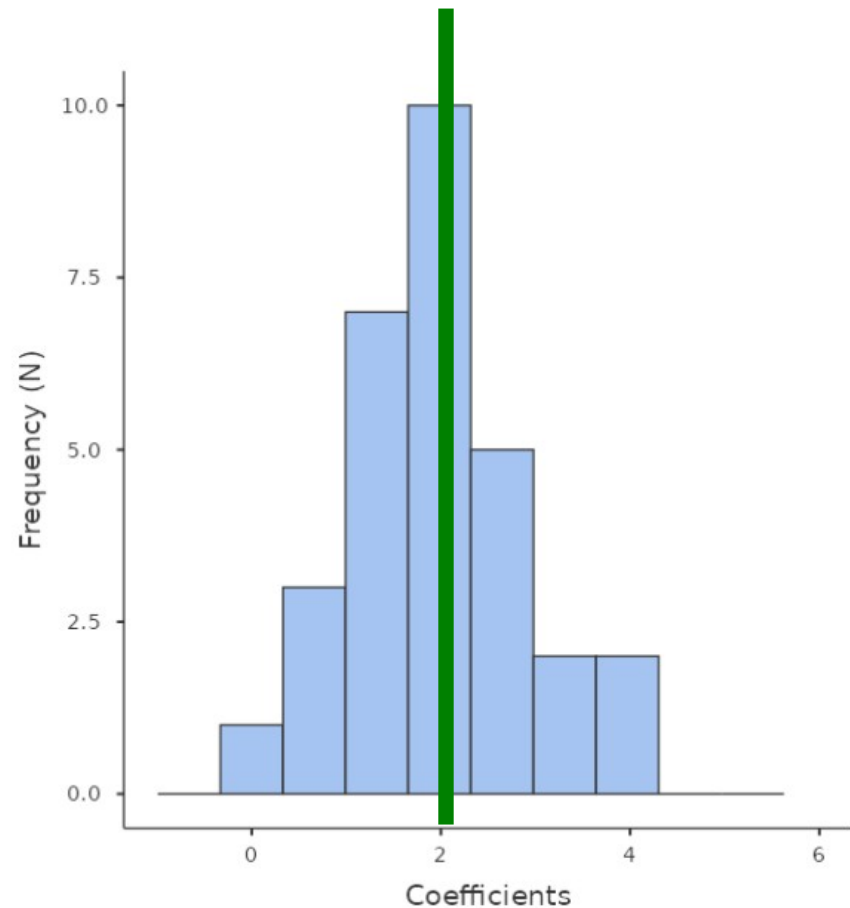
Fixed coefficients

- If coefficients vary as a variable in the population, they will have a mean and a variance, that we can estimate in our data

Mean

$$\bar{b} = \frac{\sum_j b_j}{k}$$

Recall the mean is a fixed parameter for a distribution, and so is the mean of the coefficients: it is a **fixed effect**



The Model

- We can now define a model with a regression for each cluster and the mean values of coefficients

One regression per cluster

$$\hat{y}_{ij} = a_j + b_j \cdot x_{ij}$$

Each coefficient is defined as the deviation from the mean coefficient

$$b'_j = b_j - \bar{b}$$

Overall model

$$\hat{y}_{ij} = a_j + b'_j \cdot x_{ij} + \bar{b} \cdot x_{ij}$$

The Model

- We can now define a model with a regression for each cluster and the mean value of coefficients

Overall model

$$\hat{y}_{ij} = a_j + b'_j \cdot x_{ij} + \bar{b} \cdot x_{ij}$$

**Random
coefficients**

Fixed coefficient

The mixed model

- The same goes for the intercepts

One regression per
cluster

$$\hat{y}_{ij} = a_j + b_j \cdot x_{ij}$$

Intercepts as deviations
from the average intercept

$$a'_j = a_j - \bar{a}$$

Overall model

$$\hat{y}_{ij} = \bar{a} + a'_j + \bar{b} \cdot x_{ij} + b'_j \cdot x_{ij}$$

The mixed model

- We can now define a model with a regression for each cluster and the mean values of coefficients

Overall model

$$\hat{y}_{ij} = \bar{a} + a'_j + \bar{b} \cdot x_{ij} + b'_j \cdot x_{ij}$$

**Random
coefficients**

Fixed coefficients

**A GLM which contains both fixed and
random effects is called a
Linear Mixed Model**

GLM as a special case

It is clear that everything we know for the GLM applies here: the GLM is in fact a special case of the LMM, where there are not random effects

LMM

$$\hat{y}_{ij} = \bar{a} + a'_j + \bar{b} \cdot x_{ij} + b'_j \cdot x_{ij}$$

GLM

$$\hat{y}_{ij} = \bar{a} + \bar{b} \cdot x_{ij}$$

The mixed model

- In practice, mixed models allow to estimate the kind of effects we can estimate with the GLM, but they allow the effects to vary across clusters.
- Effects that vary across clusters are called **random effects**
- Effects that do not vary (the ones that are the same across clusters) are said to be **fixed effects**

The mixed model

- To specify a correct model, we only need to understand if there are **clusters of cases** (measures or subjects) and decide which coefficients (intercepts or b coefficients) may vary across those clusters
- The fixed effects of the model are interpreted like in the GLM (regression/ANOVA)
- **Random effects** are generally not interpreted, but we can look at their variance to decide to keep them as random (variance > 0) or fix them.
- In this way we take into the account the dependence among data

Building a model

To build a model in a simple way, we need to answer very few questions:

- What is (are) the cluster variable(s)?
- What are the fixed effects?
- What are the random effects?

A clustering variable

- **What is (are) the cluster variable(s)?**
- What are the fixed effects?
- What are the random effects?
 - Any variable that groups observations (cases or measurements) such that scores may be more similar within each group than across groups
 - Any variable whose levels (groups) are a sample of a larger population of levels (groups)
 - Example: bars created groups of scores (participants) that may be more similar within the bar than across bars

Fixed effects

- What is (are) the cluster variable(s)?
- **What are the fixed effects?**
- What are the random effects?
 - Any effect that we are interested in on average (as in a standard ANOVA/Regression)
 - Example: the effect of beer on smiles in general

Fixed effects

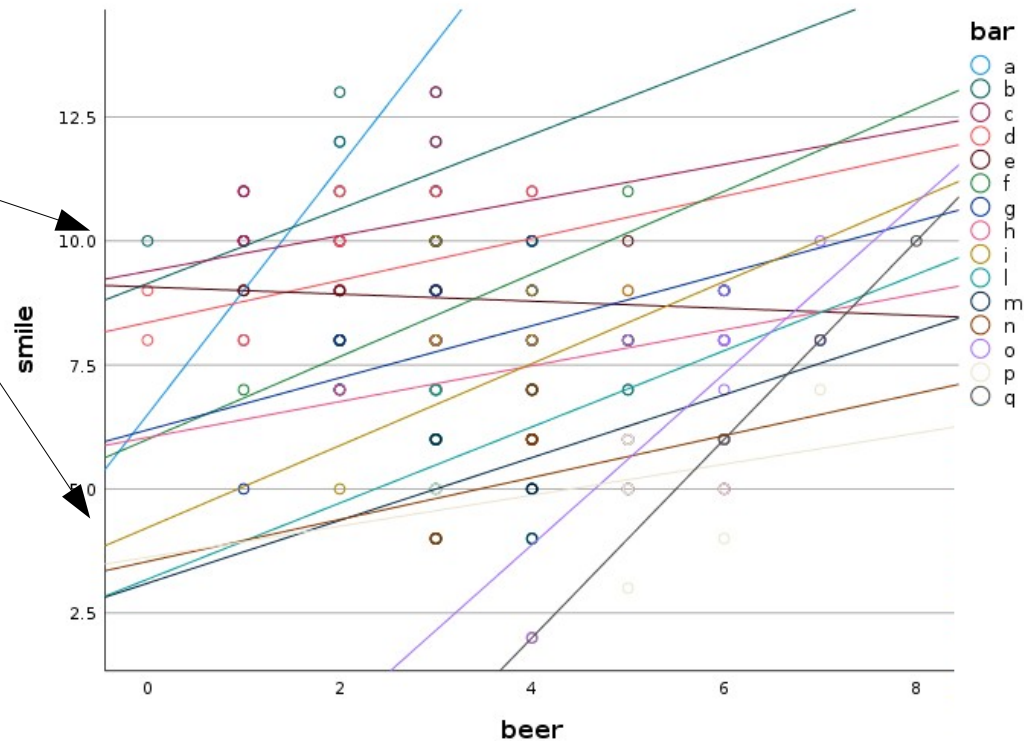
- What is (are) the cluster variable(s)?
- What are the fixed effects?
- **What are the random effects?**
 - Any effect that may vary from cluster to cluster
 - (Thus:) **Any effect that can be computed within each cluster**
 - Example: the intercepts and the effect of beer on smiles each bar

Beers at the bar

We start with a simple model

Bar

Intercepts seem to
be different from
bar to bar



Beers at the bar

We define a model where each cluster is allow to have a different intercept, the rest of the model is like a standard regression

$$y_{ij} = \bar{a} + a_j + \bar{b} \cdot x_{ij} + e_{ij}$$

- Fixed effects? Intercept and beer effect
- Random effects? Intercepts
- Clusters? bar

Authors and books may call this model:

Random-intercepts regression

or

Intercepts-as-outcomes model

Software

SPSS



R



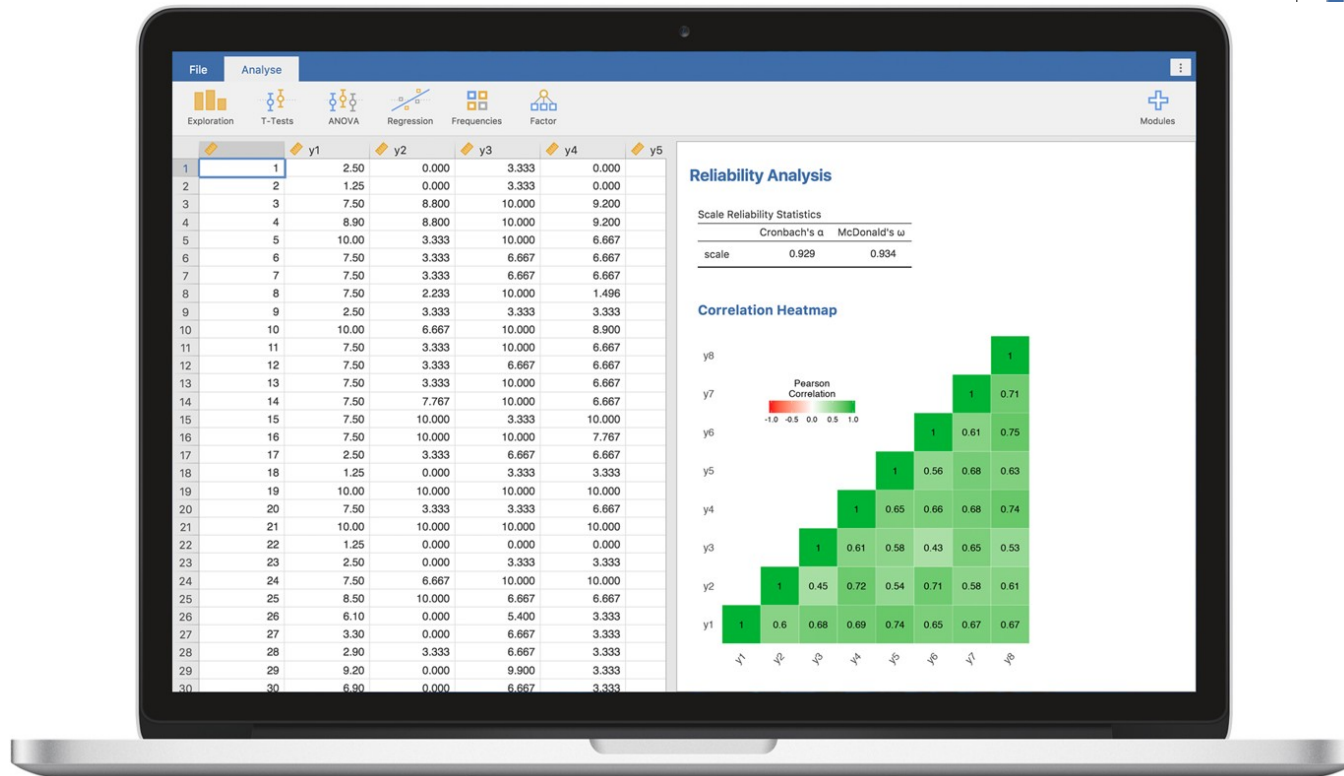
jamovi Stats.
Open.
Now.



Jamovi

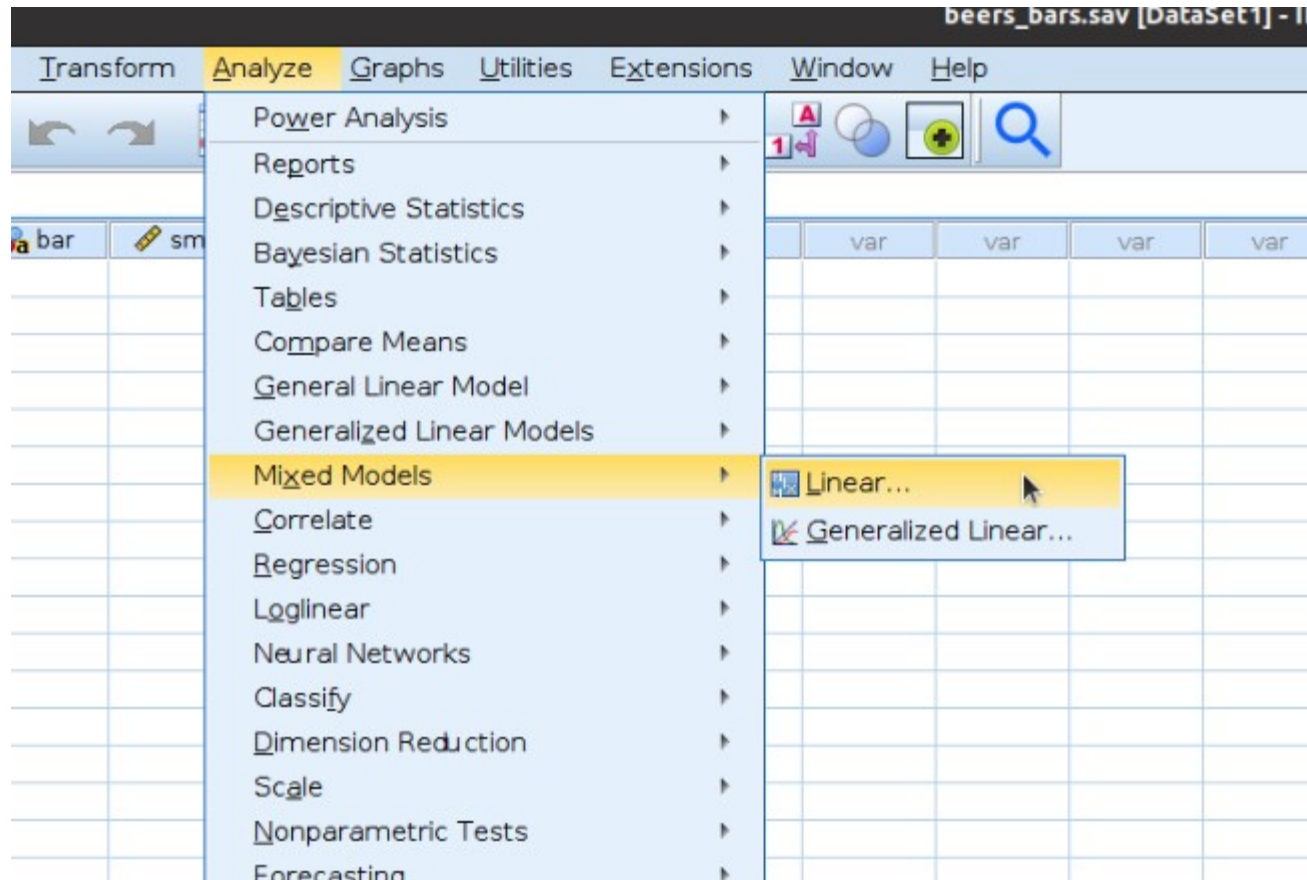
www.jamovi.org

iamovi Stats.
Open.
Now.



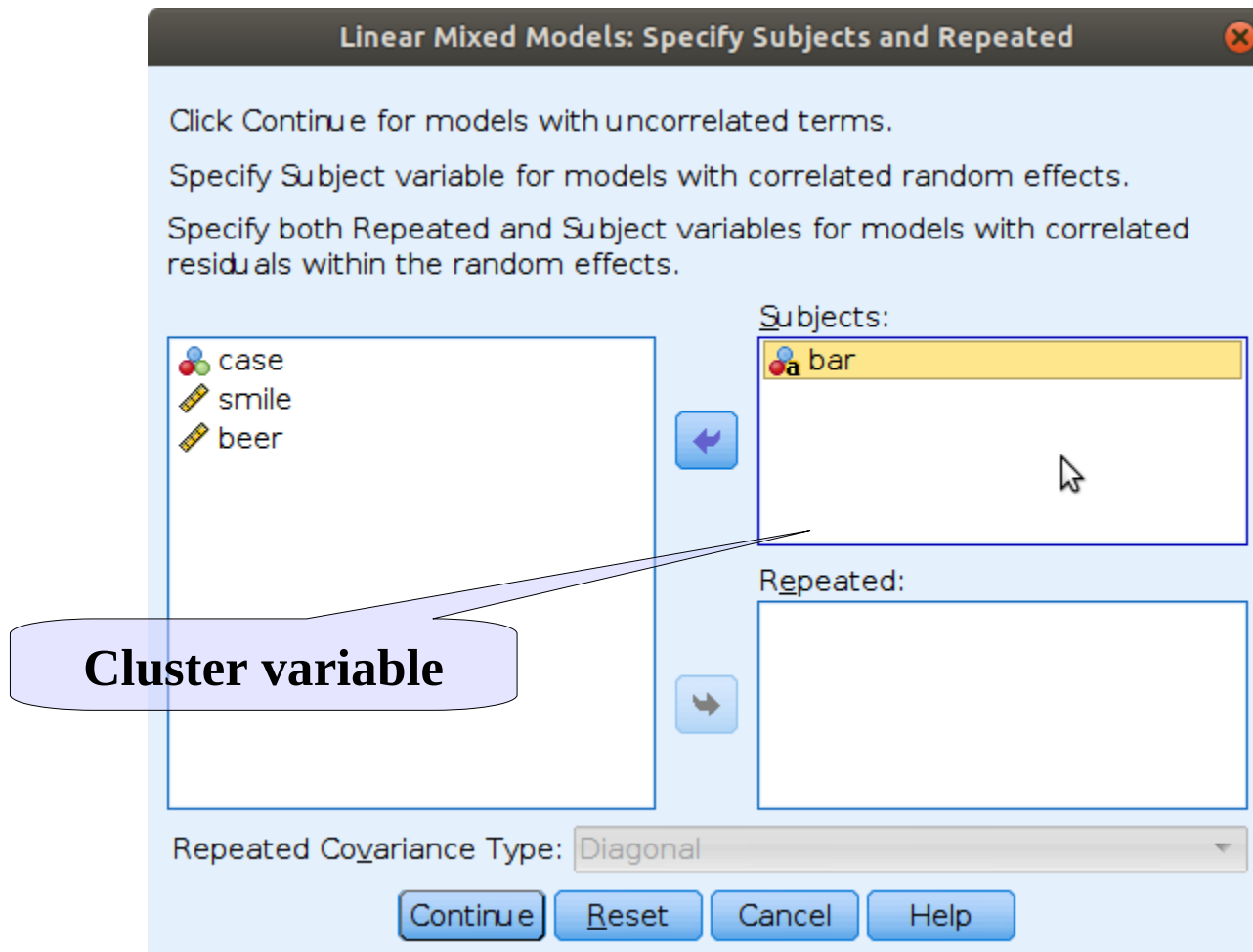
SPSS Input

Analyze → Mixed Models → Linear



SPSS Input

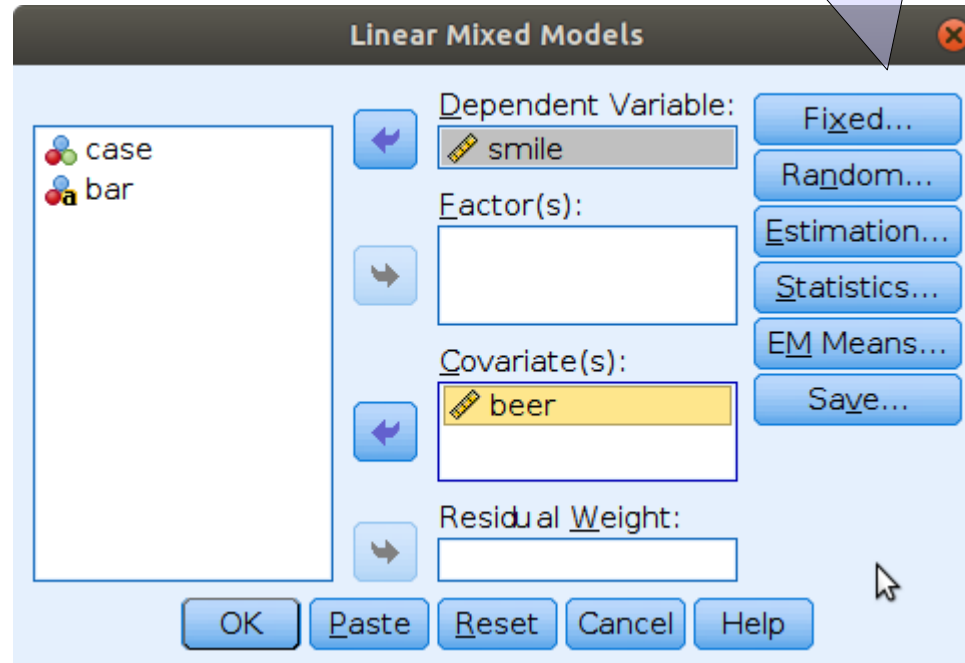
Analyze → Mixed Models → Linear



SPSS Input

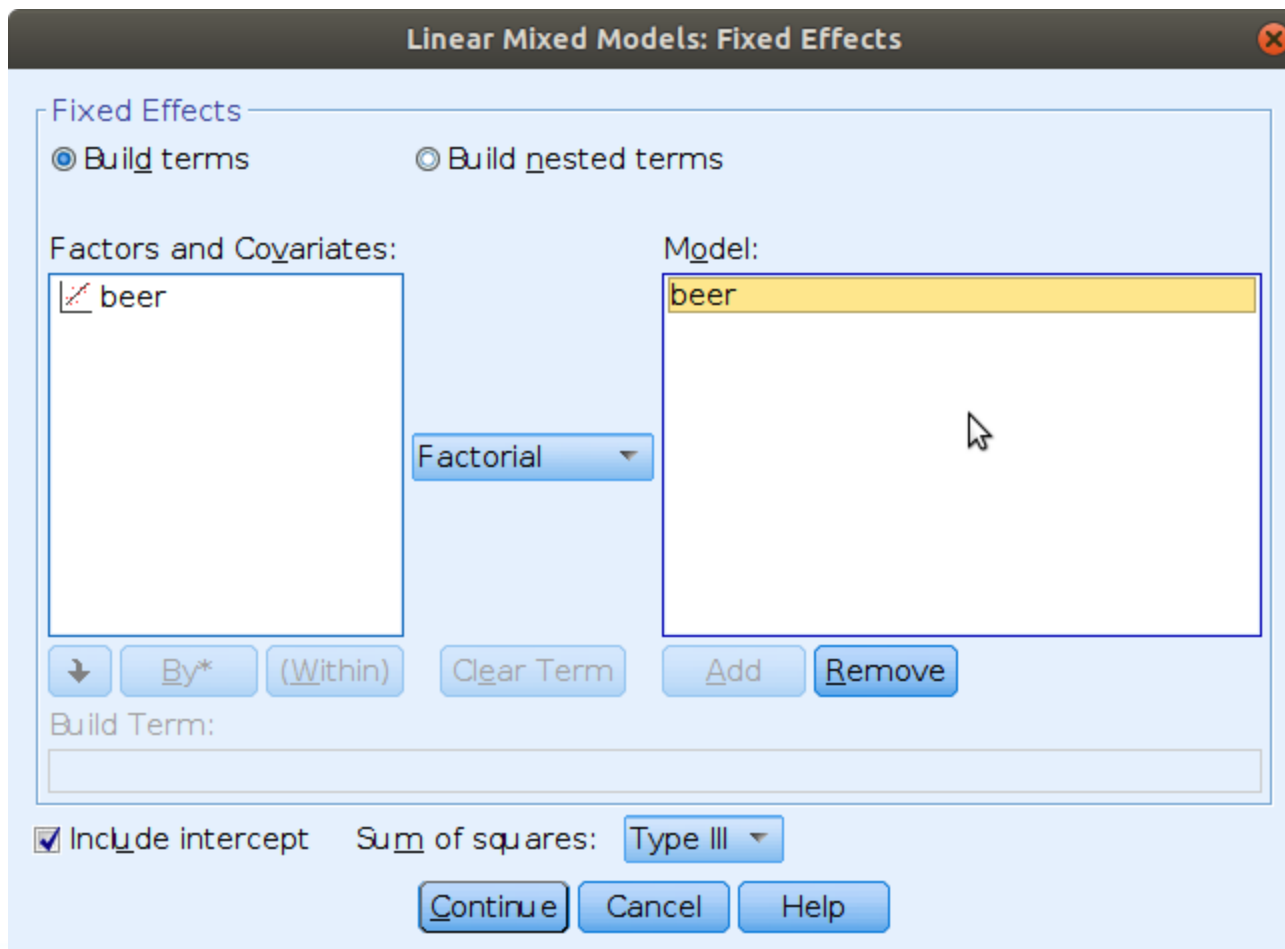
Analyze → Mixed Models → Linear

Then select Fixed



SPSS Input

Analyze → Mixed Models → Linear



The image shows the 'Linear Mixed Models: Fixed Effects' dialog box in SPSS. The window title is 'Linear Mixed Models: Fixed Effects'. Under the 'Fixed Effects' section, the 'Build terms' radio button is selected. The 'Factors and Covariates' list contains 'beer'. The 'Model' list also contains 'beer'. A 'Factorial' dropdown menu is located between the two lists. At the bottom, the 'Include intercept' checkbox is checked, and the 'Sum of squares' is set to 'Type III'. The 'Continue', 'Cancel', and 'Help' buttons are at the bottom right.

Linear Mixed Models: Fixed Effects

Fixed Effects

☒ Build terms ☐ Build nested terms

Factors and Covariates:

beer

Model:

beer

Factorial

↓ By* (Within) Clear Term Add Remove

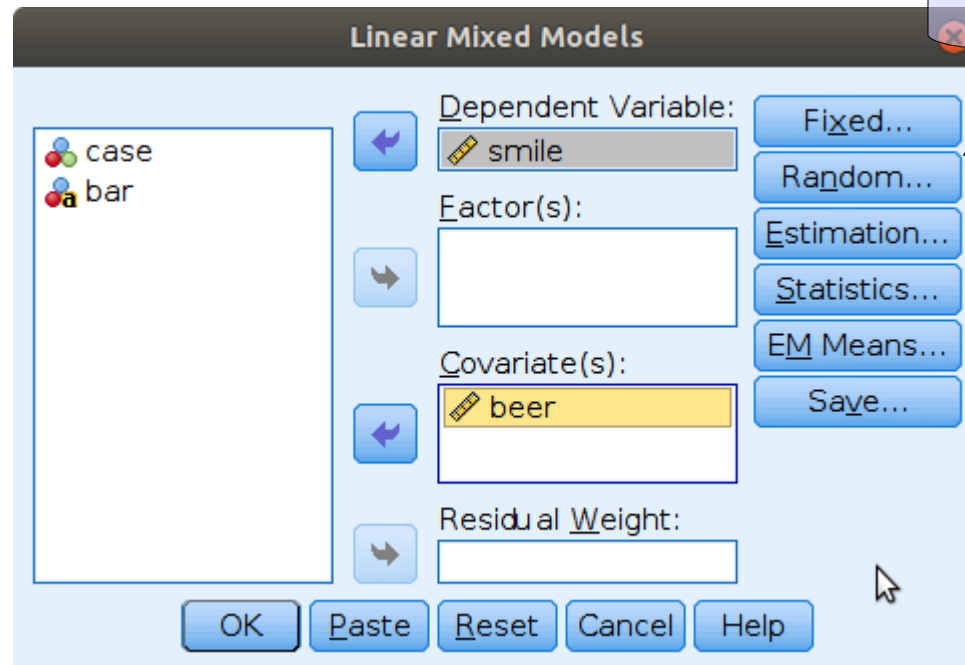
Build Term:

☒ Include intercept Sum of squares: Type III

Continue Cancel Help

SPSS Input

Analyze → Mixed Models → Linear



**Then select
Random**

SPSS Input

Analyze → Mixed Models → Linear

Linear Mixed Models: Random Effects

Random Effect 1 of 1

Previous Next

Covariance Type: Variance Components

Random Effects

☒ Build terms ☐ Build nested terms ☒ Include intercept

Factors and Covariates: Model:

beer

Factorial

↓ By* (Within) Clear Term Add Remove

Build Term:

Subject Groupings

Subjects: Combinations:

a bar

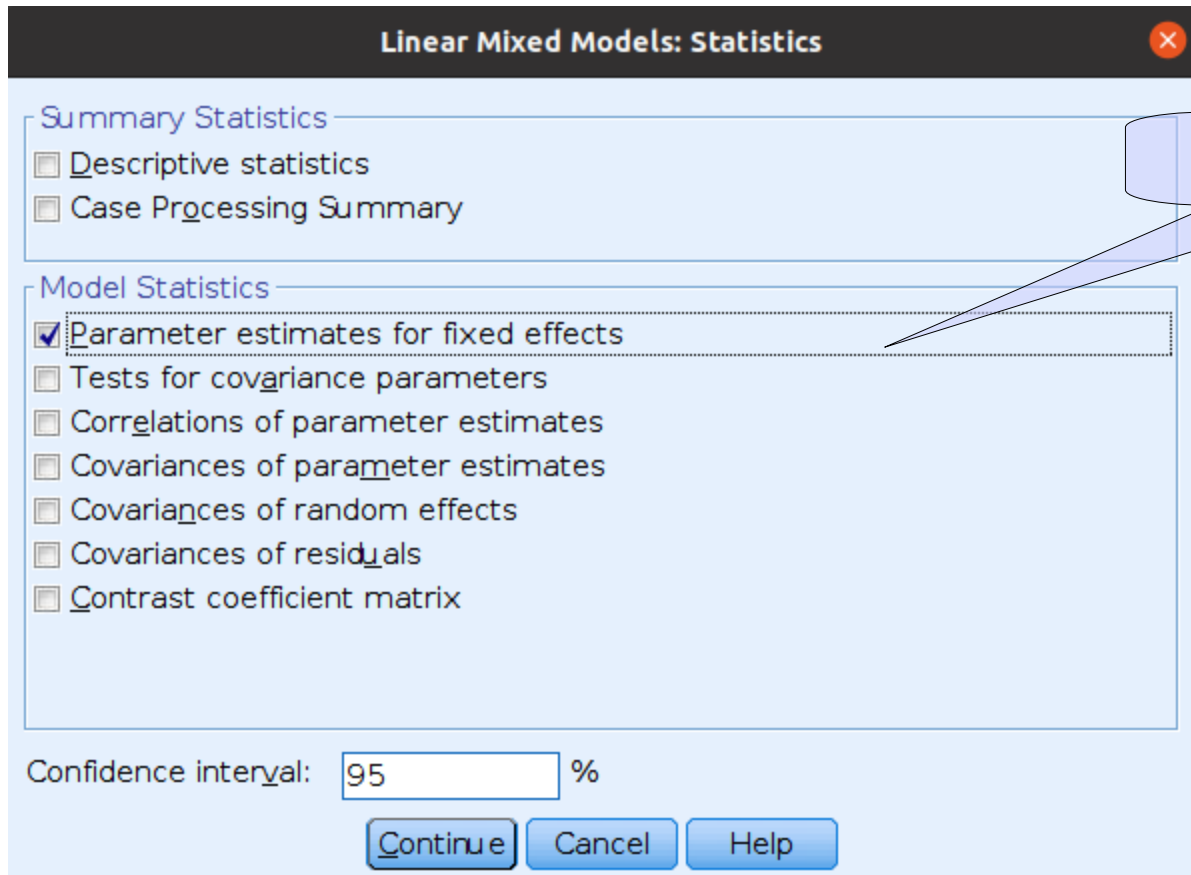
a bar

Continue Cancel Help

Random Intercept

SPSS Input

Analyze → Mixed Models → Linear



SPSS syntax

$$y_{ij} = \bar{a} + a_j + \bar{b} \cdot x_{ij} + e_{ij}$$

MIXED smile WITH beer by bar

/CRITERIA=CIN(95) MXITER(100) MXSTEP(10) SCORING(1) SINGULAR(0.000000000001) HCONVERGE(0,
ABSOLUTE) LCONVERGE(0, ABSOLUTE) PCONVERGE(0.000001, ABSOLUTE)

/FIXED=beer| SSTYPE(3)

/METHOD=REML

/print solution TESTCOV

/random intercept | SUBJECT(bar) COVTYPE(un).

Fixed effects (intercept is included by default)

Random effects

Cluster variable

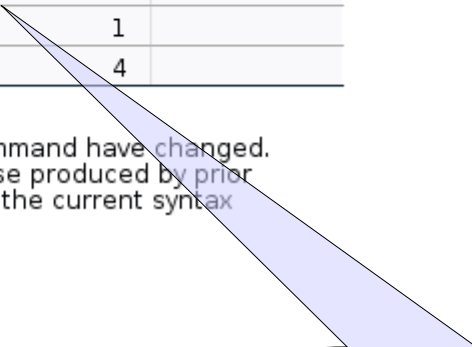
SPSS Output

Let's see if the model is how intended

		Model Dimension ^a			
		Number of Levels	Covariance Structure	Number of Parameters	Subject Variables
Fixed Effects	Intercept	1		1	
	beer	1		1	
Random Effects	Intercept ^b	1	Variance Components	1	bar
Residual				1	
Total		3		4	

a. Dependent Variable: smile.

b. As of version 11.5, the syntax rules for the RANDOM subcommand have changed. Your command syntax may yield results that differ from those produced by prior versions. If you are using version 11 syntax, please consult the current syntax reference guide for more information.



OK!

SPSS Output

We then check the variability of the random effects. If there is variability across bars, it means we were right to model the coefficients as random

Covariance Parameters

Estimates of Covariance Parameters^a

Parameter	Estimate	Std. Error
Residual	1.451189	.139257
Intercept [subject = bar] Variance	5.816514	2.320621

a. Dependent Variable: smile.

Variance greater than 0

SPSS Output

If everything is fine, we interpret the fixed effects as in any other GLM (regression)

Estimates of Fixed Effects^a

Parameter	Estimate	Std. Error	df	t	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Intercept	5.841778	.695573	19.205	8.399	<.001	4.386978	7.296578
beer	.552973	.080740	229.072	6.849	<.001	.393884	.712061

a. Dependent Variable: smile.

Intercept: On average, for zero beers we expect 5.8 smiles

SPSS Output

If everything is fine, we interpret the fixed effects as in any other GLM (regression)

Estimates of Fixed Effects^a

Parameter	Estimate	Std. Error	df	t	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Intercept	5.841778	.695573	19.205	8.399	<.001	4.386978	7.296578
beer	.552973	.080740	229.072	6.849	<.001	.393884	.712061

a. Dependent Variable: smile.

Intercept: On average, as beers increase on 1 unit, we expect smile to increase of .552 smiles

SPSS Output

We also get the overall omnibus tests

Fixed Effects

Type III Tests of Fixed Effects^a

Source	Numerator df	Denominator df	F	Sig.
Intercept	1	19.205	70.535	<.001
beer	1	229.072	46.906	<.001

a. Dependent Variable: smile.

**These are equivalent to the GLM
F-tests**

R syntax

$$y_{ij} = \bar{a} + a_j + \bar{b} \cdot x_{ij} + e_{ij}$$

```
random_beers.R* x
Source on Save
1 setwd("/home/marcello/Skinner/Teaching/Phd/MIB")
2 library(lme4)
3 library(lmerTest)
4 library(foreign)
5 dat<-read.spss('data/regression_beers_bars.sav',to.data.frame = T)
6 head(dat)
7 mm1<-lmer(smile~1+beer+(1|bar),data=dat)
8 summary(mm1)
9 |
```

Load the required libraries

Random effects

Cluster variable

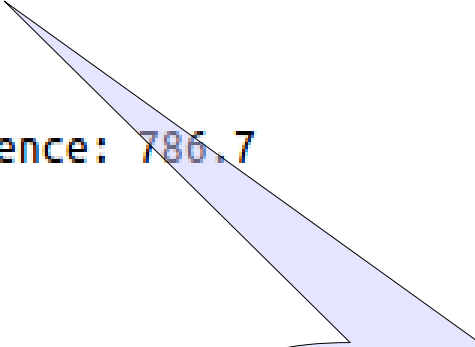
Fixed effects (intercept can be omitted as it is included by default)

R Output

Let's see if the model is how it was intended

```
> summary(mm1)
Linear mixed model fit by REML ['merModLmerTest']
Formula: smile ~ 1 + (1 | bar) + beer
Data: dat

REML criterion at convergence: 786.7
```



OK!

R Output

We then check the variability of the random effects. If there is variability across bars, it means we were right to model the coefficients as random

```
Random effects:
```

Groups	Name	Variance	Std.Dev.
bar	(Intercept)	5.817	2.412
Residual		1.451	1.205

Number of obs: 234, groups: bar, 15

Variance greater than 0

R Output

If you need a test for the random variances (Likelihood Ratio Test) run this:

```
dat<-read.spss('data/regression_beers_bars.sav',to.data.frame = T)
head(dat)
mm1<-lmer(smile~1+beer+(1|bar),data=dat)

rand(mm1)
```

ANOVA-like table for random-effects: Single term deletions

Model:

smile ~ beer + (1 | bar)

	npars	logLik	AIC	LRT	Df	Pr(>Chisq)
<none>	4	-405.86	819.72			
(1 bar)	3	-495.44	996.89	179.17	1	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

OK!

R Output

If everything is fine, we interpret the fixed effects as in any other GLM (regression)

```
Fixed effects:
              Estimate Std. Error      df t value Pr(>|t|)
(Intercept)   5.84178     0.69557   19.20481    8.399 7.45e-08 ***
beer          0.55297     0.08074  229.07224    6.849 6.78e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Intercept: On average, for zero
beers we expect 5.5 smiles**

Note that without `library(lmerTest)` you do not get the p.values!

R Output

If you prefer the F-test, use `anova()`

Default DF

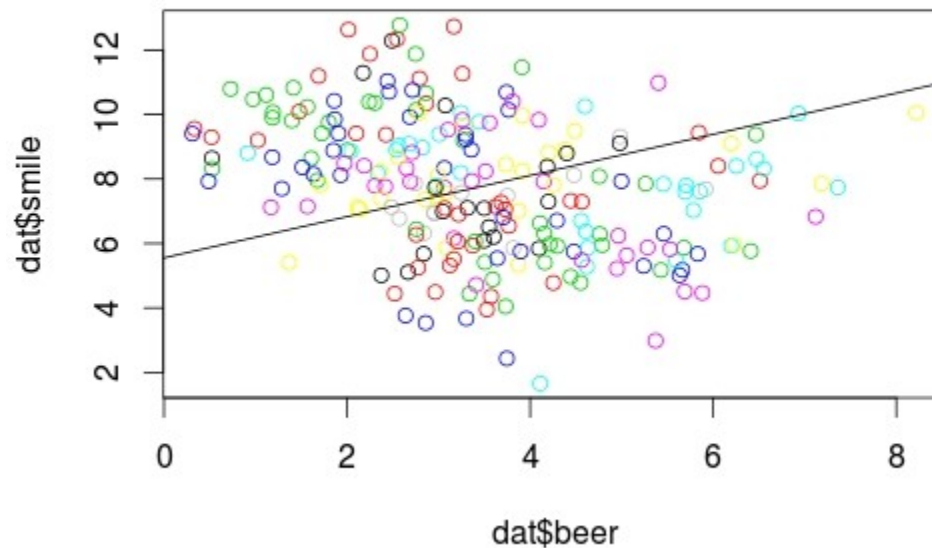
```
-----  
Type III Analysis of Variance Table with Satterthwaite's method  
      Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)  
beer  68.069   68.069     1 229.07  46.906 6.783e-11 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

R Output

Plot fixed effects:

```
2 summary(mm1)
3 plot(dat$smile~dat$beer,col=dat$bar)
4 abline(fixef(mm1))
5
```

fixef extracts fixed coefficients from the model

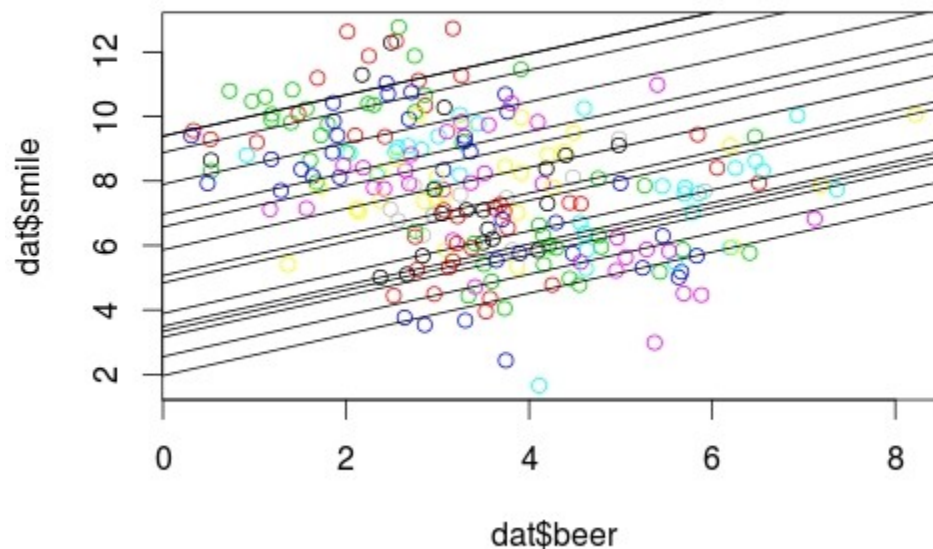


R Output

Plot random effects effects:

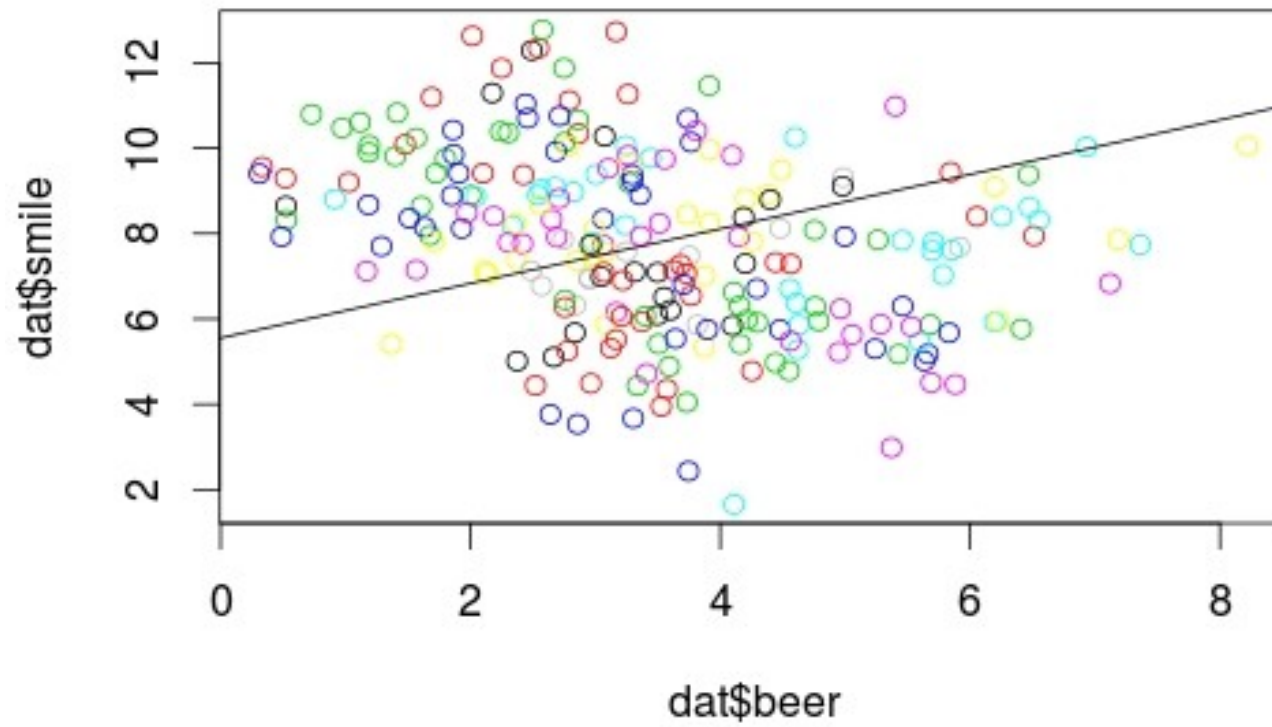
```
16  
17 plot(dat$smile~dat$beer,col=dat$bar)  
18 apply(coef(mm1)[[1]],1,abline)  
19 |
```

coef extracts random coefficients from the model



Plots

- Plotting fixed effects is simply plotting the straight line as in any linear model



Jamovi

- In jamovi mixed models can be estimated with the GAMLi module

The screenshot displays the Jamovi software interface with the 'Analyses' tab selected. The 'Mixed Model' module is active, showing a list of variables (A, case, smile, beer, bar) on the left. The 'Dependent Variable' field is empty, and the 'Factors' field is also empty. The 'Covariates' and 'Cluster variables' fields are empty. The 'Estimation' section shows 'REML' selected, and 'Confidence Intervals' are set to 95%. The 'Fixed Effects' section is expanded, showing a table with columns: F, Num df, Den df, p. The 'Random Effects' section is also expanded, showing a table with columns: Groups, Name, SD, Variance. A callout bubble points to the 'Fixed Effects' section with the text: 'All options are in expandable panels'.

Mixed Model

Model Info

Info

Get started Select the dependent variable

Get started Select at least one cluster variable

Get started Select at least one term in Random Effects

Optional Select factors and covariates

Fixed Effect ANOVA

F	Num df	Den df	p
---	--------	--------	---

Fixed Effects Parameter Estimates

Effect	Contrast	Estimate	SE	Lower	Upper	df	t	p
--------	----------	----------	----	-------	-------	----	---	---

Random Components

Groups	Name	SD	Variance
--------	------	----	----------

Estimation Confidence Intervals

☒ REML ☒ Confidence Intervals Interval 95 %

> Fixed Effects

> Random Effects

> Factors Coding

> Covariates Scaling

> Post Hoc Tests

> Fixed Effects Plots

> Simple Effects

> Estimated Marginal Means

Jamovi

- In jamovi mixed models can be estimated with the GAMLj module

Mixed Model

Variables: A, case

Dependent Variable: smile

Factors:

Covariates: beer

Cluster variables: bar

Estimation: ☒ REML

Confidence Intervals: ☒ Confidence intervals Interval: 95 %

Define the variables role

Jamovi

- In jamovi mixed models can be estimated with the GAMLj module

Define the fixed effects

The screenshot shows the 'Fixed Effects' dialog box in the Jamovi GAMLj module. It features two main panels: 'Components' on the left and 'Model Terms' on the right. The 'Components' panel contains a list with the item 'beer'. Between the panels are two buttons: a right-pointing arrow and a right-pointing arrow with a dropdown arrow. The 'Model Terms' panel contains a list with the item 'beer'. At the bottom left, there is a checked checkbox labeled 'Fixed Intercept'.

Fixed Effects

Components

beer

Model Terms

beer

☒ Fixed Intercept

Jamovi

- In jamovi mixed models can be estimated with the GAMLj module

Define the random component

The screenshot shows the 'Random Effects' dialog box in the Jamovi GAMLj module. The dialog is divided into several sections:

- Components:** A list box containing 'Intercept | bar' and 'beer | bar'. The 'beer | bar' component is selected and highlighted in blue. A right-pointing arrow button is located between this list and the 'Random Coefficients' list.
- Random Coefficients:** A list box containing 'Intercept | bar'.
- List components:** A section with a checked checkbox for 'Model terms' and an 'Add' button next to a dropdown menu currently set to 'None'.
- Effects correlation:** A section with three radio button options: 'Correlated' (which is selected), 'Not correlated', and 'Correlated by block'.
- Tests:** A section with a checked checkbox for 'LRT for Random Effects'.

- As soon as you define the random component, you get the results

Model Results

Model Fit

Type	R ²	df	LRT X ²	p
Conditional	0.818	2	201.438	< .001
Marginal	0.090	1	38.630	< .001

[4]

>

R-squared Marginal: How much variance can the fixed effects alone explain of the overall variance

R-squared Conditional: How much variance can the fixed and random effects together explain of the overall variance

- As soon as you define the random component, you get the results

F-test for the main effect of beer

Model Results

Fixed Effect Omnibus tests

	F	Num df	Den df	p
beer	46.906	1	229.072	< .001

Note. Satterthwaite method for degrees of freedom

Jamovi

- As soon as you define the random component, you get the results

**SPSS intercept was different: GAMLj
centers the variables by default**

**coefficients for the main effect of
beer**

Fixed Effects Parameter Estimates

Names	Estimate	SE	95% Confidence Interval		df	t	p
			Lower	Upper			
(Intercept)	7.770	0.630	6.535	9.005	13.166	12.330	< .001
beer	0.553	0.081	0.395	0.711	229.072	6.849	< .001

- As soon as you define the random component, you get the results

Random Components

Groups	Name	SD	Variance	ICC
bar	(Intercept)	2.412	5.817	0.800
	Residual	1.205	1.451	

Note. Number of Obs: 234 , groups: bar 15

Random components

ICC

- The intra-class correlation indicates the proportion of variance explained by the differences across clusters

**ICC=Intra-class
correlation**

Random Components				
Groups	Name	SD	Variance	ICC
bar	(Intercept)	2.412	5.817	0.800
	Residual	1.205	1.451	

Note. Number of Obs: 234 , groups: bar 15

$$ICC = \frac{\sigma_a}{\sigma_a + \sigma}$$

$$ICC = \frac{5.817}{5.817 + 1.451} = .8$$

- Jamovi can plot the results

Plots

Horizontal axis
→ beer

Separate lines
→

Separate plots
→

Display

☒ None
☐ Confidence intervals
☐ Standard Error

Plot

☐ Observed scores
☐ Y-axis observed range

Use

☐ Original scale
☐ Varying line types

Random Effects

☒ Random effects
☒ Average method
☐ Full predicted method

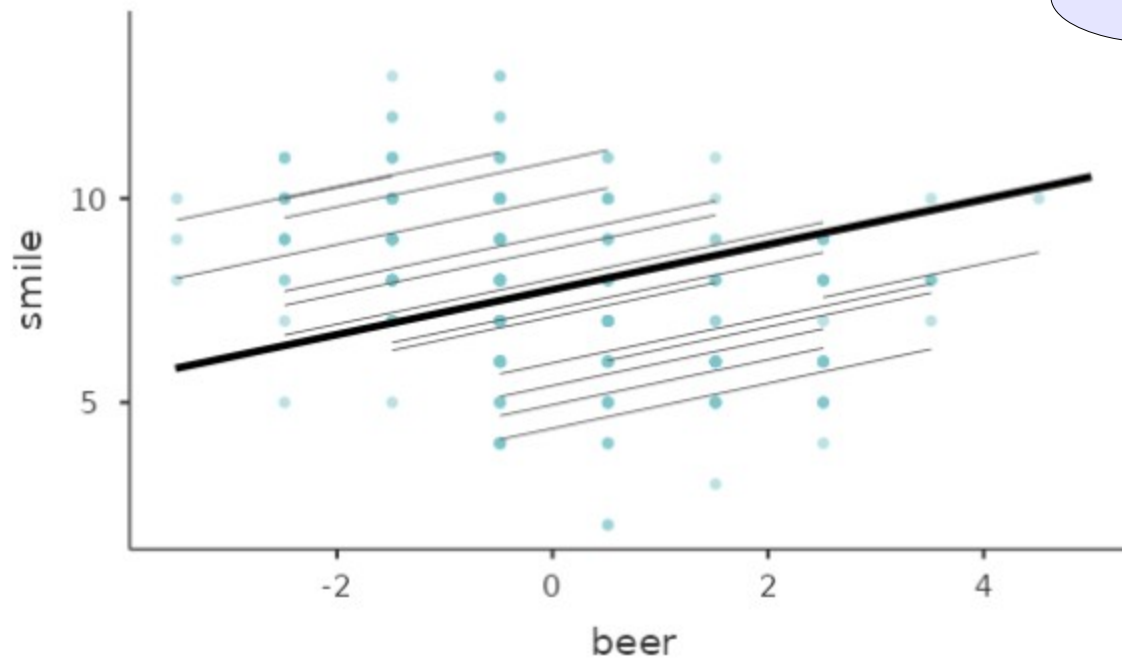
Plot (here just one line)

Jamovi

- Jamovi can plot fixed and random effects

Fixed + Random effects

Effects Plots



● Jamovi test for random effects variances

Random Effects

Components

Intercept | bar

beer | bar

Random Coefficients

Intercept | bar

List components

☒ Model terms

Add

None

Effects correlation

☒ Correlated
 ☐ Not correlated
 ☐ Correlated by block

Tests

☒ LRT for Random Effects

Random Effect LRT

Test	N. par	AIC	LRT	df	p
(1 bar)	3.00	997	179	1.00	< .001

Beers at the bar 2

We can now try a model where also the **b** coefficients are allow to vary across clusters

$$y_{ij} = \bar{a} + a_j + \bar{b} \cdot x_{ij} + b \cdot x_{ij} + e_{ij}$$

- Fixed effects? Intercept and beer effect
- Random effects? Intercepts and b coefficients
- Clusters? bar

Some authors may call this model:
Random-coefficients regression
or
Intercepts- and Slopes-as-outcomes model

Jamovi

- We keep the same setup for the variables

Mixed Model

Dependent Variable

→ smile

Factors

→

Covariates

→ beer

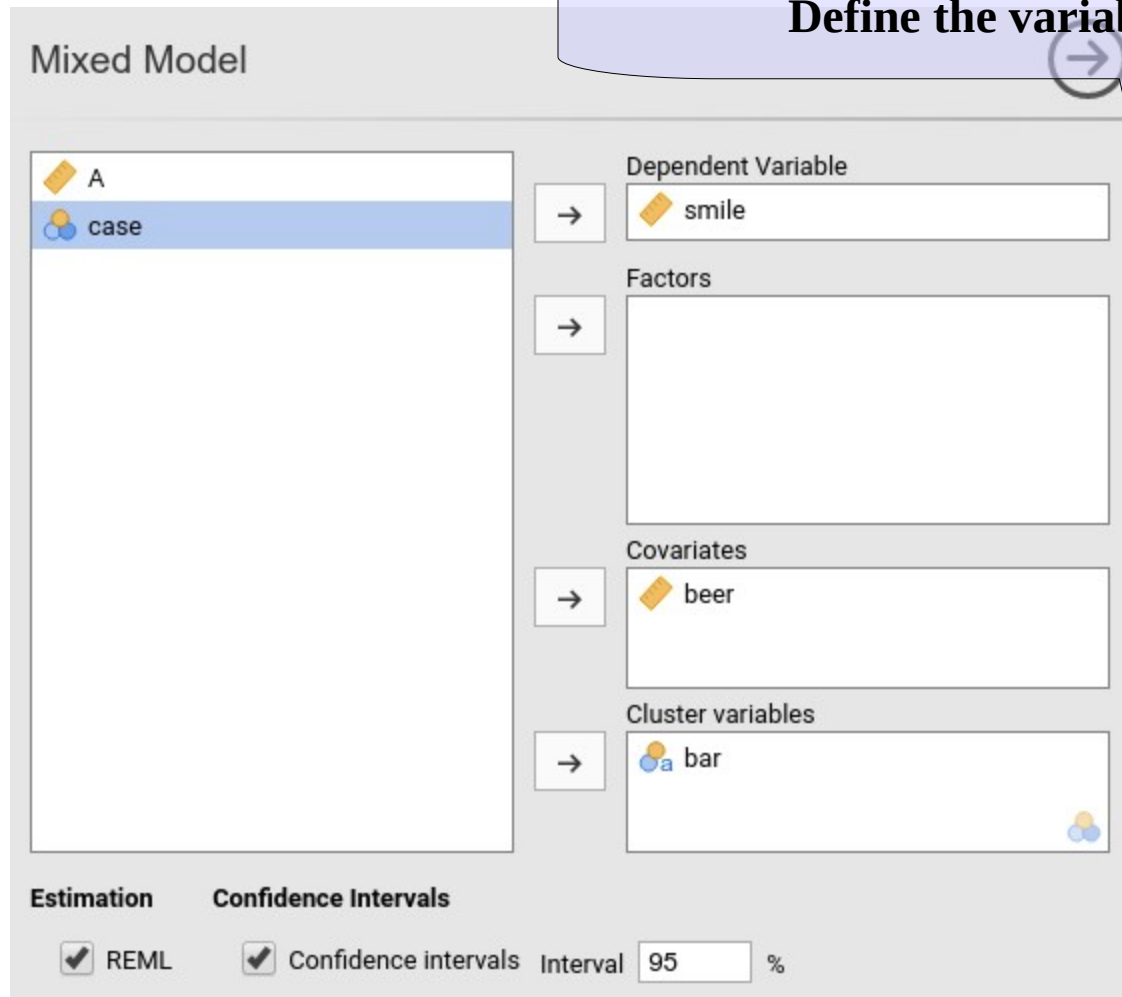
Cluster variables

→ bar

Estimation

Confidence Intervals

☒ REML ☒ Confidence intervals Interval 95 %



Define the variables role

- We add the effect of beer as a random coefficient

Define the random component

The screenshot shows the 'Random Effects' dialog box in Jamovi. The 'Components' section is empty. The 'Random Coefficients' section contains two entries: 'Intercept | bar' and 'beer | bar'. The 'Effects correlation' section has three radio buttons: 'Correlated' (selected), 'Not correlated', and 'Correlated by block'. The 'Tests' section has a checkbox for 'LRT for Random Effects' which is unchecked.

Random Effects

Components

Random Coefficients

Intercept | bar
beer | bar

Effects correlation

Correlated
Not correlated
Correlated by block

Tests

LRT for Random Effects

- As soon as you define the random component, you get the results

Model Results

Model Fit

Type	R ²	df	LRT X ²	p
Conditional	0.822	4	203.003	< .001
Marginal	0.090	1	17.016	< .001

[4]

>

R-squared Marginal: How much variance can the fixed effects alone explain of the overall variance

R-squared Conditional: How much variance can the fixed and random effects together explain of the overall variance

- As soon as you define the random component, you get the results

Model Results

F-test for the main effect of beer

Fixed Effect Omnibus tests

	F	Num df	Den df	p
beer	36.057	1	7.234	< .001

Note. Satterthwaite method for degrees of freedom

- As soon as you define the random component, you get the results

**coefficients for the main effect of
beer**

Fixed Effects Parameter Estimates

Names	Estimate	SE	95% Confidence Interval		df	t	p
			Lower	Upper			
(Intercept)	7.610	0.633	6.368	8.851	12.928	12.013	< .001
beer	0.555	0.093	0.374	0.737	7.234	6.005	< .001

- As soon as you define the random component, you get the results

Random Components

Groups	Name	SD	Variance	ICC
bar	(Intercept)	2.417	5.842	0.803
	beer	0.167	0.028	
Residual		1.196	1.431	

Note. Number of Obs: 234 , groups: bar 15

Random Parameters correlations

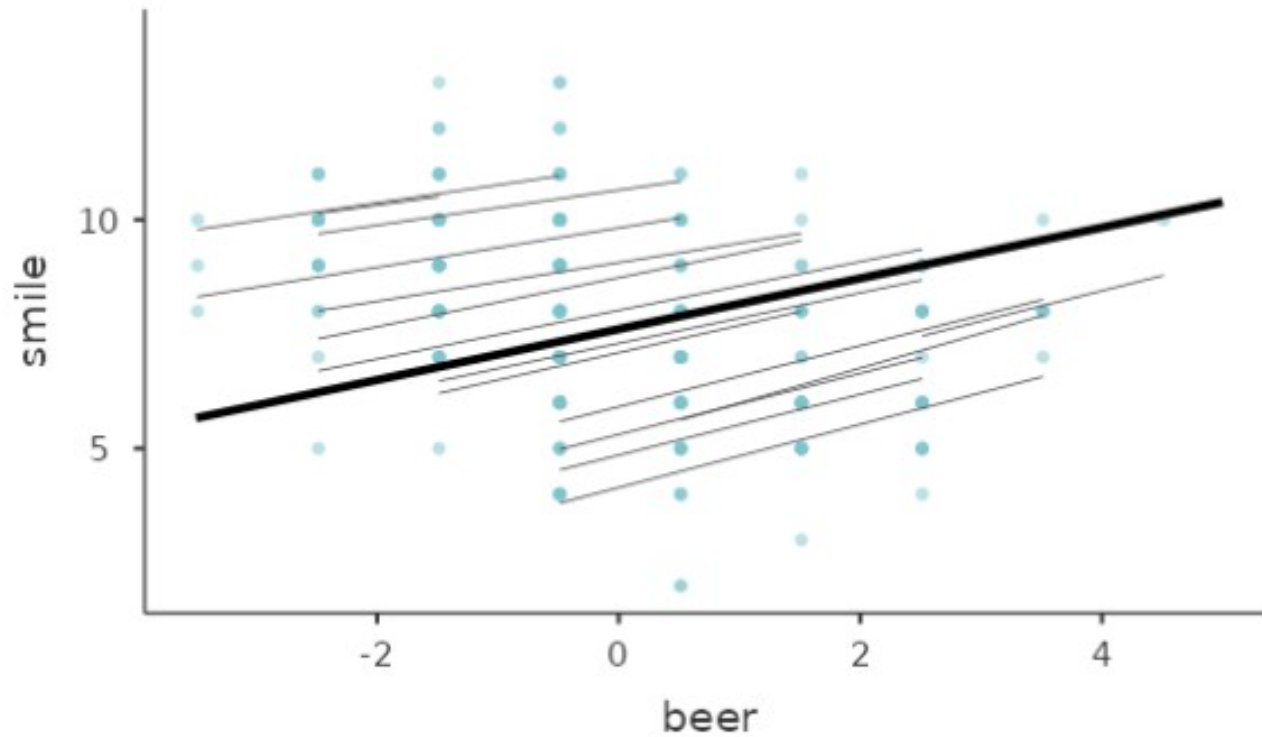
Groups	Param.1	Param.2	Corr.
bar	(Intercept)	beer	-0.766

**Random coefficients
variances**

**Random coefficients
correlation**

- In this case is fixed and random effects regression lines

Effects Plots

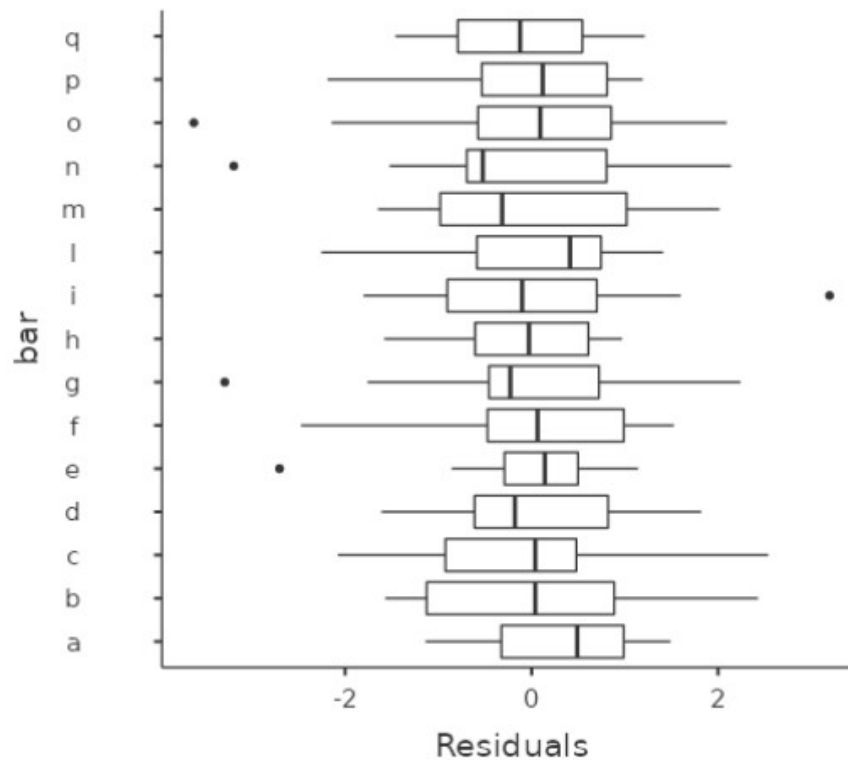


Residuals plot

- We can also look at the residual plot by cluster

Residuals by cluster boxplot

Clustering variable: bar



Mixed Linear Models

- With the mixed model one can take into the account dependency among measures (within clusters) almost in any situation
- It allows applying the GLM logic to a broader range of designs
- Any kind of independent variables
- Efficient handling of missing values
- **Multi-level research designs**
- Repeated measures designs
- Generalizes to the generalized linear model (logistic etc)

Multi-level models

Multi-level models

- The Multi-level model is **not** a “statistical technique”!
- The Multi-level model is **an approach** to analyze multi-level designs
- **The multi-level model is estimated using a mixed model**
- What is peculiar:
 - The importance of the clustering variables (higher levels)
 - The research questions
 - The cluster level is called group level (*group=cluster in this terminology*)

Example

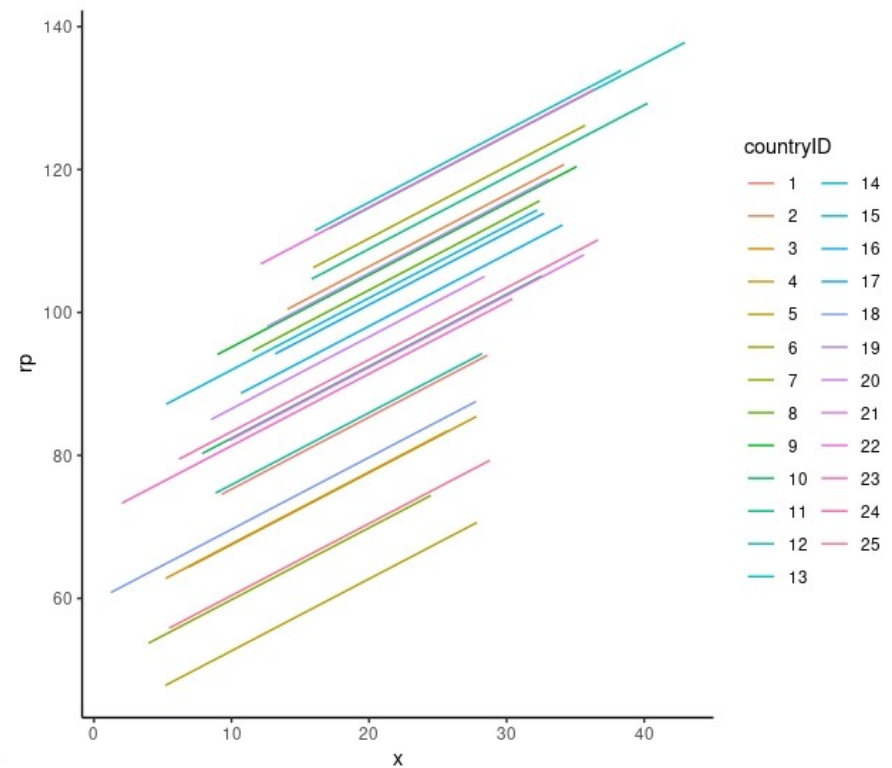
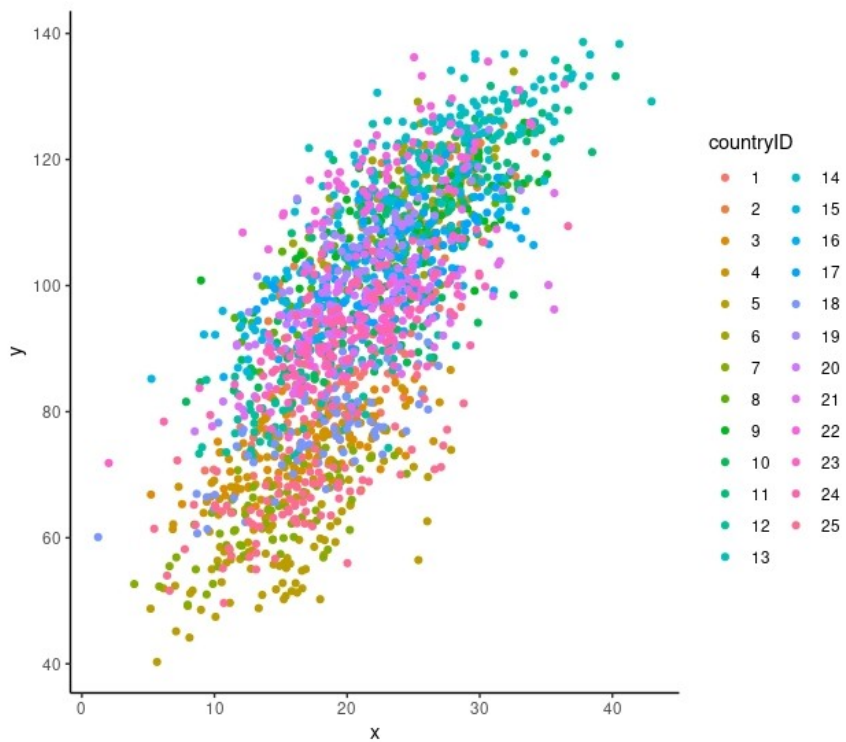
- Assume we have a multi-country research, in which we measured individuals (people) *charity contribution* and their *income (individual level)*.
- We can have information about country taxes regulations (*country level*)
- We are interested on the relationship between *contribution* and *income* at the **individual level and at the country level**

Questions

- We are interested on the relationship between cooperation and income at the **individual levels and at the country level**
- Independently of the country: Does people with higher income contribute more? (*individual level effect*)
- Independently of the people: Do countries with average higher income show higher average contribution? (*country level effect*)

Structure vs aim

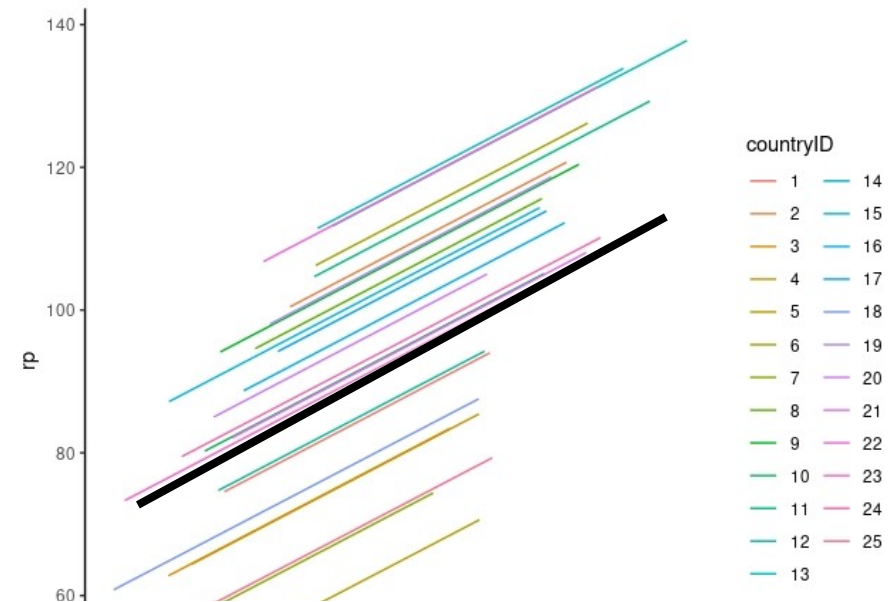
- If we only look at the data structure, we are in the beers at bars example
- But the research aim is different



Individual level

- If we fit a model like *beer at bars*, we only get the individual level effect, averaged across countries

Independently of the country: Does people with higher income contribute more? (*individual level effect*)



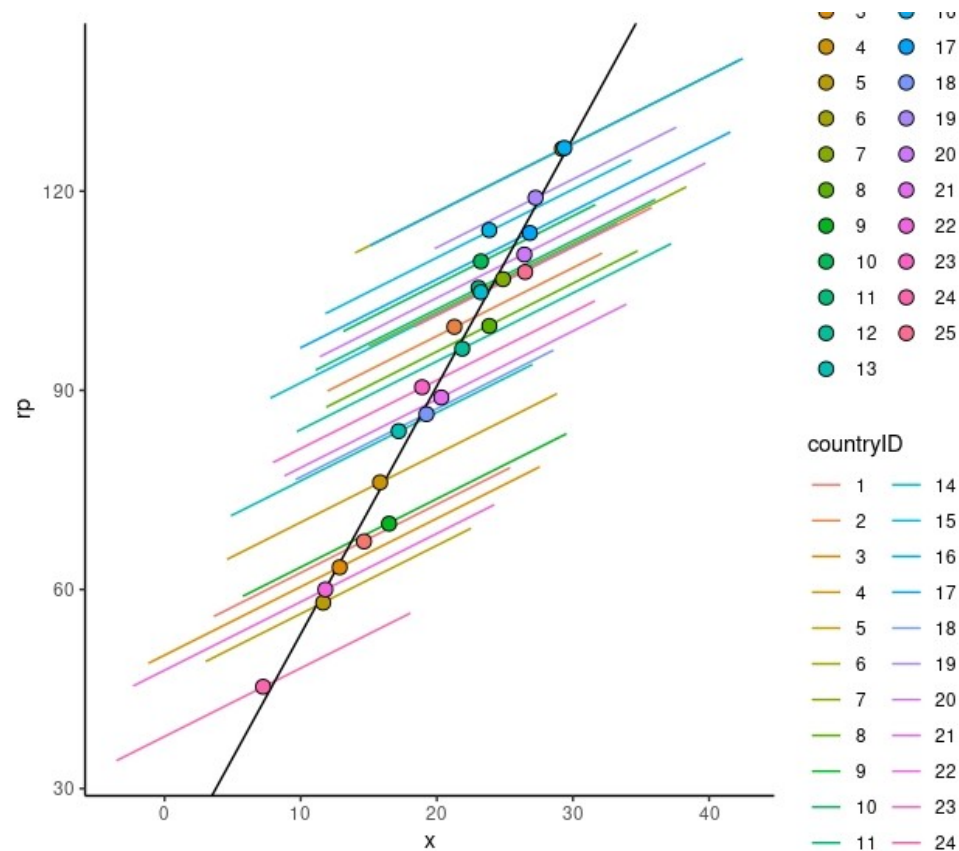
Fixed Effects Parameter Estimates

Names	Estimate	SE	95% Confidence Interval		df	t	p
			Lower	Upper			
(Intercept)	95.73	3.0129	89.830	101.64	24.0	31.8	< .001
x	1.01	0.0235	0.964	1.06	1928.6	43.0	< .001

Country level

- But income may have an effect also at the country (second) level

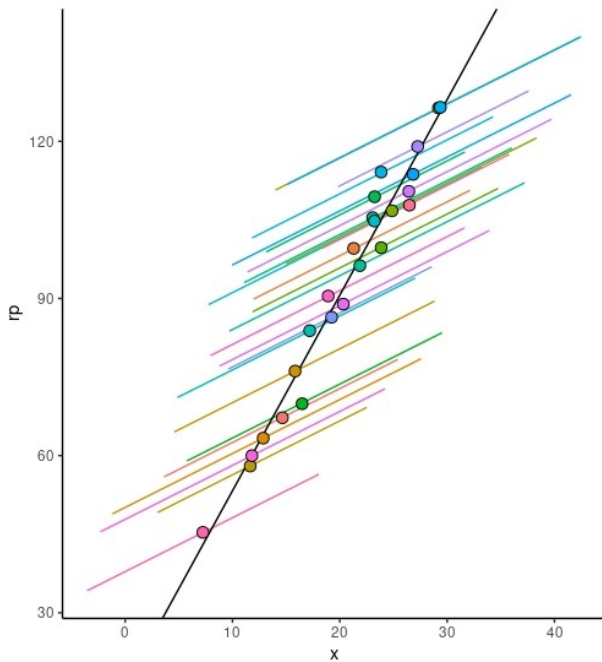
Independently of the people: Do countries with average higher income show higher average contribution? (*country level effect*)



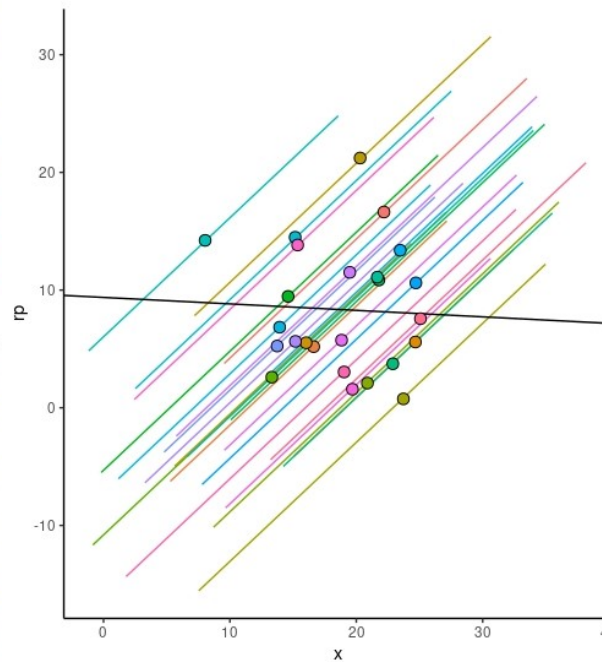
Country vs individual level

- The effect of a variable at each level is **independent** of the effect at any other level

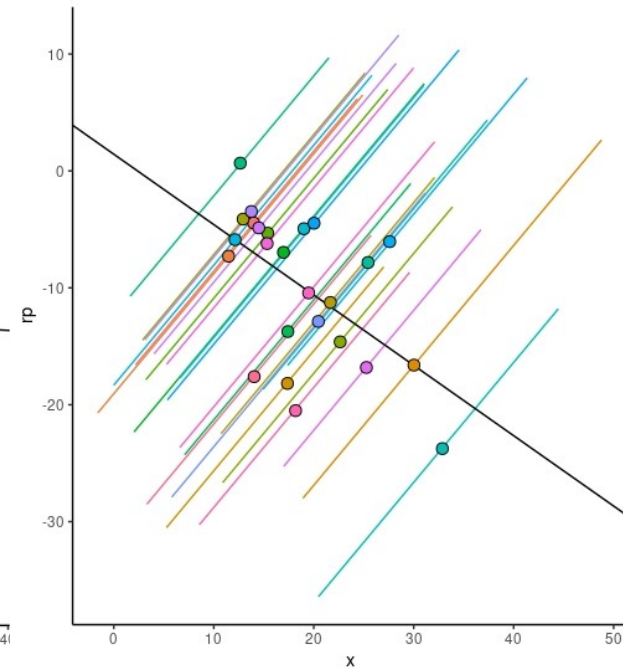
Individual +, country +



individual +, country 0



individual +, country -

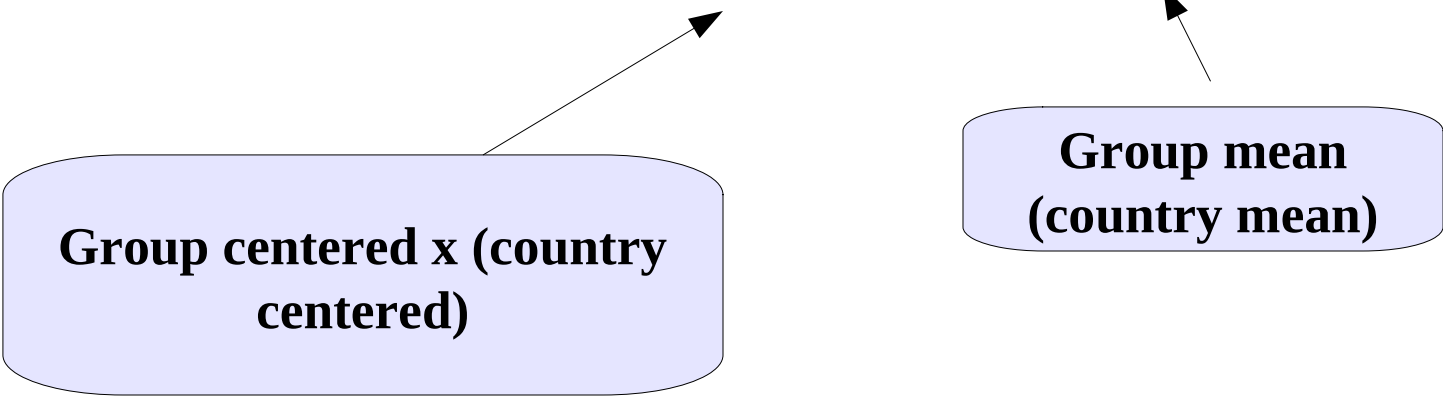


The mixed model

- To capture the effect of countries (second level) we should include the country levels means
- To make it independent of people levels, we group-center individual level x

$$\hat{y}_{ij} = \bar{a} + a'_j + b_1 \cdot (x_{ij} - \bar{x}_j) + b_2 \cdot \bar{x}_j$$

Group centered x (country centered)



**Group mean
(country mean)**

Coefficients

- The model returns the effects at level 1 (individuals) and level 2 (country)

$$\hat{y}_{ij} = \bar{a} + a'_j + b_1 \cdot (x_{ij} - \bar{x}_j) + b_2 \cdot \bar{x}_j$$

Independently of the country: Does people with higher income contribute more?

Do countries with average higher income show higher average contribution?

Data

- We simply compute two new variables: the group centered x and the group means

$$xcen = (x_{ij} - \bar{x}_j)$$

$$xm = \bar{x}_j$$

countryID	x	y	xm	xcen
10	12.919	88.400	20.547	-7.628
10	27.128	98.083	20.547	6.581
10	21.709	99.612	20.547	1.163
10	20.586	94.523	20.547	0.040
10	14.580	81.559	20.547	-5.967
10	28.697	106.248	20.547	8.150
10	8.937	84.683	20.547	-11.609
10	17.241	94.658	20.547	-3.306
10	18.252	84.677	20.547	-2.295
10	18.913	90.324	20.547	-1.633
11	33.202	129.347	28.333	4.870
11	23.598	114.079	28.333	-4.735
11	23.746	109.661	28.333	-4.587
11	26.870	113.275	28.333	-1.463
11	25.305	116.171	28.333	-3.028
11	31.789	114.460	28.333	3.456
11	18.956	119.912	28.333	-9.377
11	32.524	123.181	28.333	4.191
11	27.646	111.706	28.333	-0.717

Model

- We use them as independent variables

Mixed Model 

 A
 x
 rp



→

Dependent Variable
 y

→

Factors

→

Covariates
 xcen
 xm

→

Cluster variables
 countryID

Random Coefficients

Intercept | countryID
xcen | countryID

Results (fixed effect)

- And interpret the coefficients accordingly

Fixed Effects Parameter Estimates

Names	Estimate	SE	95% Confidence Interval		df	t	p
			Lower	Upper			
(Intercept)	95.65	1.3013	93.104	98.21	23.1	73.5	< .001
xcen	1.01	0.0278	0.954	1.06	21.8	36.3	< .001
xm	4.37	0.3126	3.757	4.98	23.1	14.0	< .001

Within each country: as people income increases of 1 unit, contribution increases of **1.01**

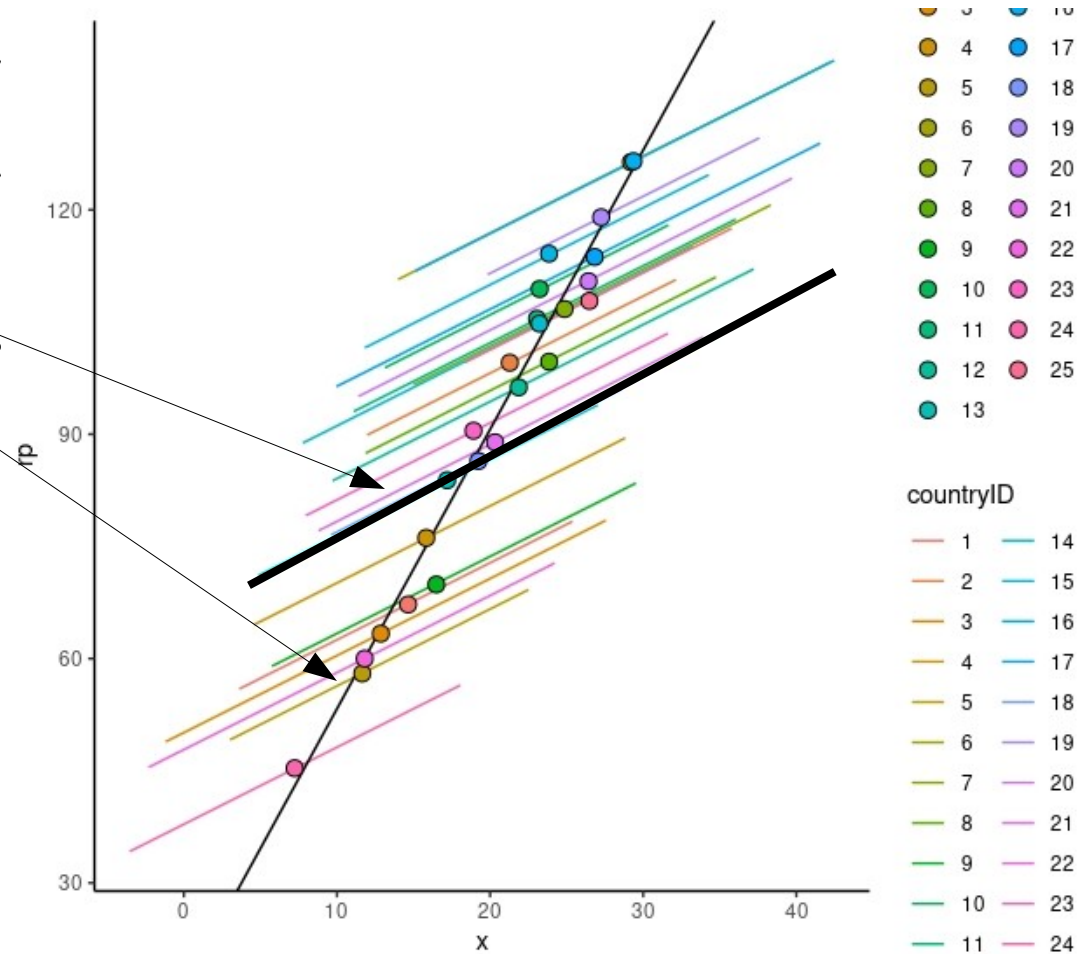
Across countries: As the average income of a country increases 1 unit, the average contribution increases of **4.37** units

Results (fixed effect)

- And interpret the coefficients accordingly

Fixed Effects Parameter Estimates

Names	Estimate	SE
(Intercept)	95.65	1.3013
xcen	1.01	0.0278
xm	4.37	0.3126



Multi-level models

- **The multi-level model is estimated using a mixed model**
- What is peculiar:
 - We want to estimate predictors effects at each level

Mixed Linear Models

- With the mixed model one can take into the account dependency among measures (within clusters) almost in any situation
- It allows applying the GLM logic to a broader range of designs
- Any kind of independent variables
- Efficient handling of missing values
- Multi-level research designs
- **Generalizes to the generalized linear model (logistic etc)**
- **Repeated measures designs**

